GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINES MARKET

Market Analysis, Insights, Trends, and Opportunity Analysis, 2020-2027

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GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINES MARKET

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• Market Size and Forecast, 2017 - 2027, (USD Million)	●	●	●
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• Market Size and Forecast, 2017 - 2027, (USD Million)	●	●	●
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**Key Pointers are applicable to other companies as well.*

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○ Coriolis Composites

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SECTION 1

GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINES MARKET

- Research Objectives And Assumptions

RESEARCH OBJECTIVES

This report analyzes the current scenario as well as the future market potential for global automated fiber placements and automated tape laying (AFP- ATL) machines market. The market for the same has been segmented on the basis of machine type, end-user, and region.

Key features of the study:

- This report provides in-depth analysis of global automated fiber placements and automated tape laying (AFP- ATL) machines market, and provides market size (US$ Mn), and compound annual growth rate (CAGR) (%) for the forecast period (2020 – 2027)
- It elucidates potential revenue opportunity across different segments and explains attractive investment proposition matrix for this market. It provides valuable insights about market drivers, restraints, opportunities, trends and competitive strategies adopted by the leading players.
- It profiles leading players in the global automated fiber placements and automated tape laying (AFP- ATL) machines market based on the following parameters – company overview, financial performance, product portfolio, and strategy. Insights from this report would allow marketers and the management authorities of the companies to make informed decisions with respect to their adoption, product launches and market expansion.
- The global automated fiber placements and automated tape laying (AFP- ATL) machines market report caters to various stakeholders in this industry, including investors, manufacturers, service providers, suppliers, and other industry participants. Companies would greatly benefit in decision-making through various strategy matrices used in analyzing the global automated fiber placements and automated tape laying (AFP- ATL) machines market.

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ASSUMPTIONS

- **Market & Forecast :**
  - Changes in the inflation rate has not been considered while forecasting market numbers
  - Macroeconomic indicators such as GDP, regulatory decisions, investments, and spending have been considered to arrive at the market estimates
  - Secondary axis represents Y-o-Y growth
  - Bottom-up approach has been followed to assess market numbers for type of segment
  - Top-down approach has been used to counter-validate the estimated numbers for each segment
- **Currency :**
  - All the values estimated are in US$ (U.S. Dollar)
- **Scope :**
  - Year of Estimation – 2020
  - Actual Year/Base Year - 2019
  - Historic Year – 2017 and 2018
  - Forecast Period – 2020 to 2027

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ABBREVIATION

CAGR	Compound Annual Growth Rate
Y-o-Y	Year-on-Year
Mn	Million
APAC	Asia Pacific
MEA	Middle East and Africa
AFP	Automated Fiber Placements
ATL	Automated Tape Laying
OEM	Original Equipment Manufacturer
NCC	National Composites Centre
CFRP	Carbon Fiber Reinforced Polymer

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SECTION 2

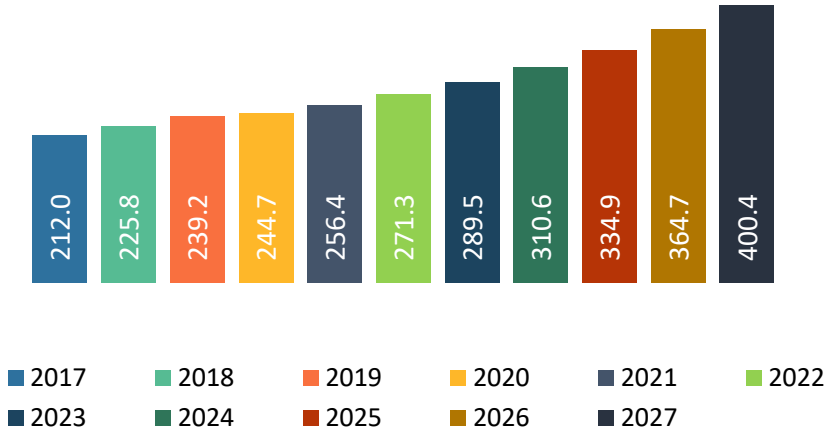
GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINES MARKET

- Market Overview

GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINES MARKET OVERVIEW

MARKET OVERVIEW

FIGURE 2.1 Global Automated Fiber Placements and Automated Tape Laying (AFP-ATL) Machines Market Value (US$ Mn) Analysis and Forecast



7.3% Projected growth of global automated fiber placements and automated tape laying (AFP- ATL) machines market revenue between 2020 and 2027

- In 2019, automated fiber placements and automated tape laying (AFP-ATL) machines market generated revenue of US$ 256.4 million and is expected to reach US$ 400.4 million by 2027, exhibiting a CAGR of 7.3% over the forecast period (2020 to 2027)
- Automated fiber placement is a composites manufacturing process of heating and compacting synthetic resin pre-impregnated non-metallic fibers on typically complex tooling mandrels. ATL machines are designed for high speed tape laying of aircraft structural components. AFP- ATL technology aids in better precision and increased deposition rates.
- The rising automation in the aerospace industry is expected to drive the global automated fiber placements and automated tape laying (AFP-ATL) machines market growth. The demand for AFP- ATL machines is rapidly growing owing to increasing need for revolutionizing the composite manufacturing process.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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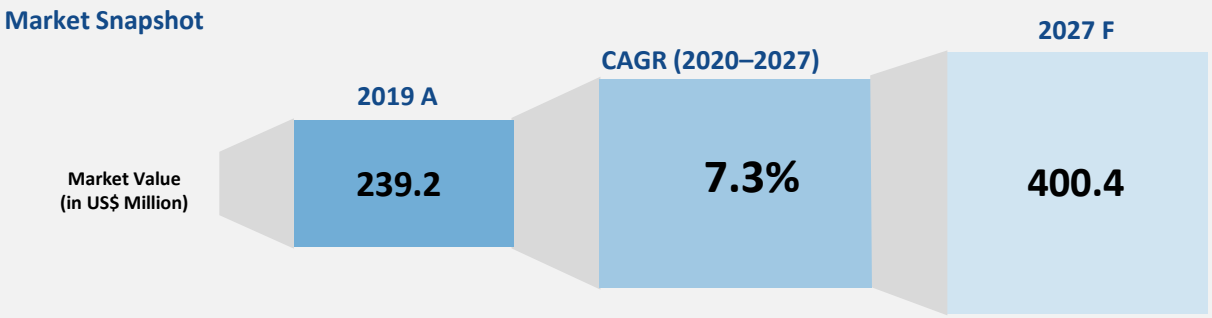
GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINES MARKET OVERVIEW



- Based on the machine type, automated fiber placements and automated tape laying (AFP- ATL) machines market is segmented into AFP Machine, ATL Machine, and Hybrid AFP/ATL Machine. AFP machines segment is expected to account for largest share in global automated fiber placements and automated tape laying (AFP- ATL) machines market over the forecast period.
- On the basis of region, the global automated fiber placements and automated tape laying (AFP- ATL) machines market is segmented into North America, Europe, Asia Pacific, Latin America, and Middle East and Africa. North America region accounted for largest share in global automated fiber placements and automated tape laying (AFP- ATL) machines market. Asia Pacific region is expected to exhibit high CAGR over the forecast period.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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MARKET SNIPPET, BY MACHINE TYPE

TABLE 2.1 Global Automated Fiber Placements and Automated Tape Laying (AFP- ATL) Machines Market Value (US$ Mn) Analysis and Forecast, By Machine Type

Machine Type	2018	2023	2027
AFP Machine	101.9	145.1	218.3
ATL Machine	115.4	131.1	164.6
Hybrid AFP/ATL Machine	8.5	13.3	17.5

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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MARKET SNIPPET, BY END USER

TABLE 2.2 Global Automated Fiber Placements and Automated Tape Laying (AFP- ATL) Machines Market Value (US$ Mn) Analysis and Forecast, By End User

End User	2018	2023	2027
Tier Players	53.6	68.6	94.8
OEMs (Original Equipment Manufacturers)	159.2	206.6	288.6
Others	13.0	14.3	18.0

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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MARKET SNIPPET, BY REGION

TABLE 2.3 Global Automated Fiber Placements and Automated Tape Laying (AFP- ATL) Machines Market Value (US$ Mn) Analysis and Forecast, By Region

Region	2018	2023	2027
North America	72.8	93.6	129.7
Europe	61.4	78.7	108.9
Asia Pacific	51.1	68.6	98.5
Latin America	19.1	23.9	32.4
Middle East	13.8	16.5	21.6
Africa	7.7	8.2	9.2

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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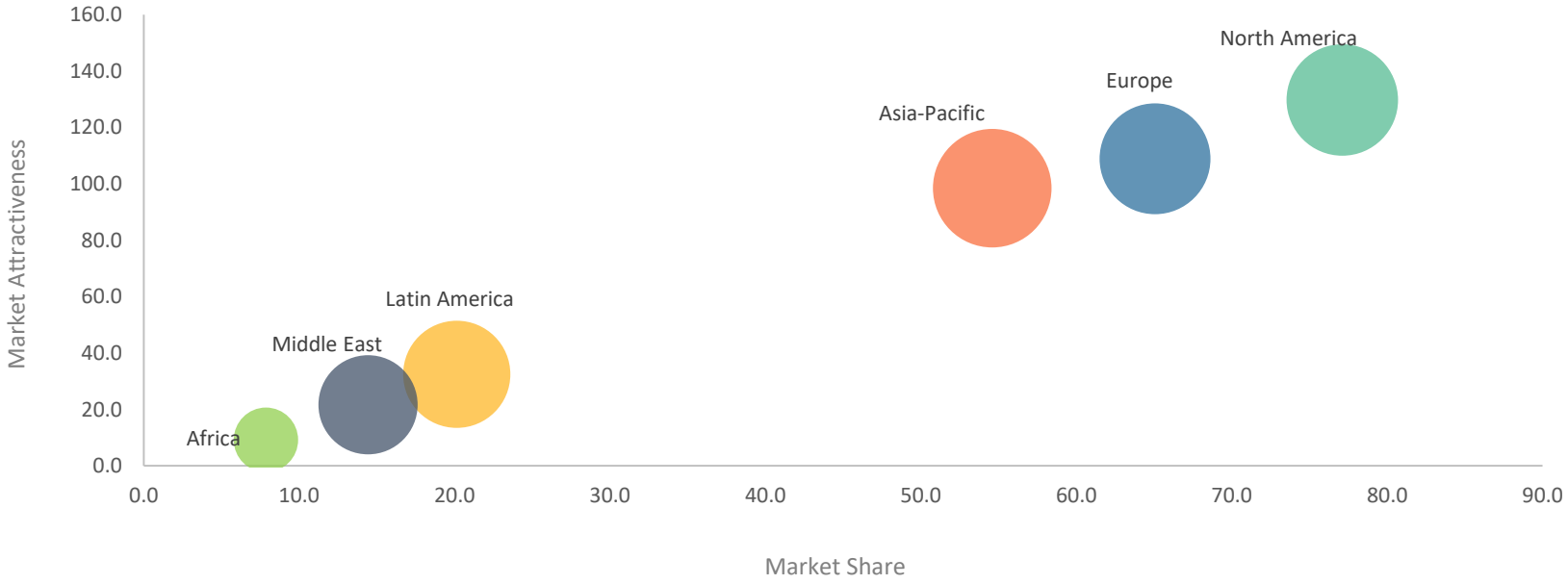
MARKET OPPORTUNITY MAP



Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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SECTION 3

GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINES MARKET

- Market Dynamics

RISING AUTOMATION IN THE AEROSPACE INDUSTRY

1

- Automation provides many benefits within the aerospace industry. Automation increases efficiency and productivity of business processes and offers quality control. Aerospace manufacturing is a specialized and complex process. Automated fiber placement and tape laying (AFP/ATL) are the technologies that have revolutionized the production of composite structures for the aviation and space industry.
- Complex composite structures that have been built manually (using hand lay-up) are automatically produced using AFP/ATL technologies, resulting in higher throughput, improved composite structure quality, reduced scrap material, reduced labor costs, reduced manufacturing time, accuracy, and repeatability of the production process.
- The comprehensive software solutions in automated fiber placement and tape laying (AFP/ATL) machines allow complex shapes to be programmed quickly
- For instance, Mikrosam offers Special AFP solution – multi-robotic work cell, which provides maximum production flexibility for the applications in aerospace industry

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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RISING NEED FOR REVOLUTIONIZING THE COMPOSITE MANUFACTURING PROCESS

2

- Automated Fiber Placement (AFP) is one of the most advanced methods for fabricating composite structures. This method is used exclusively with continuous fiber reinforced tape.
- Automated fiber placement machine uses a robot to automatically place material and build a structure one ply (layer) at a time. AFP machines decrease production time per part without having to increase individual machine performance. This enables the creation of higher production rate cells without the need to develop higher performing individual machines and components. For components used in the aerospace industries, where a high level of quality is required, the automated tape laying (ATL) machines reduce layup errors and material waste.
- For instance, MTorres group offers TORRESFIBERLAYUP, an automated fiber placement (AFP) machine designed to provide the highest productivity when it comes to fabricating high contour carbon fiber aircraft components

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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HIGH CAPITAL COST OF AFP-ATL MACHINES

- The automated fiber placements and automated tape laying (AFP- ATL) machines are expensive and need high capital investment. This is one of the primary restraining factors for growth of the automated fiber placements and automated tape laying (AFP- ATL) machines market. Also, the automated fiber placements and automated tape laying AFP-ATL machines are very complex and need skilled laborers to operate it.
- The high cost of automated fiber placements and automated tape laying (AFP-ATL) machines along with the need for highly skilled technicians are key factors that limit its usage in the highly cost-sensitive markets such as automotive and building & construction.
- For instance, in 2013, National Composites Centre (NCC) invested US$ 4 million in composite manufacturing equipment- automated fiber placement (AFP) machine in Bristol, England

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINES MARKET DYNAMICS (IMPACT ANALYSIS)

COHERENT MARKET INSIGHTS

GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINES MARKET, DRIVERS AND RESTRAINTS IMPACT ANALYSIS, 2020-2027

	DRIVERS	1-2 Years	3-5 Years	6-8 Years
MARKET DRIVERS	RISING AUTOMATION IN THE AEROSPACE INDUSTRY	●●●○	●●●○	●●●●
	RISING NEED FOR REVOLUTIONIZING THE COMPOSITE MANUFACTURING PROCESS	●●●●	●●●○	●●●○
MARKET RESTRAINTS	HIGH CAPITAL COST OF AFP-ATL MACHINES	●●●○	●●●○	●●○○

●●●● High Impact ●●●○ Medium Impact ●●○○ Low Impact

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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INCREASING APPLICATION AREAS OF AFP MACHINES IN AUTOMOTIVE, WIND ENERGY, AND MARINE INDUSTRIES

- Automated fiber placement (AFP) technology is used mainly in the aerospace industry. However, automotive, wind energy, and marine are likely to be new focus areas for the market participants in the near future as these industries are increasingly working on the development of manufacturing processes that efficaciously address their complex work areas related to prepreg part manufacturing.
- Additionally, manufacturers are focusing on offering Automated Fiber Placement (AFP) machines for manufacturing high scale, high performance components in wind energy and automotive industries. Automated Tape Layup (ATL) or Automated Fiber Placement (AFP) machines are used in the manufacturing of wind turbine blades to reduce labor cost and improve quality.
- For instance, Coriolis Composites, a leader in the supply of automated fiber placement machines, implemented its solutions for a number of industrial sectors with a strong track record in aeronautics and automotive. The Coriolis Composites offers thermoplastic fiber placement machine for manufacturing automotive structural composite parts.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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DEVELOPMENTS TO IMPROVE THE PERFORMANCE AND RELIABILITY OF AUTOMATED FIBER PLACEMENT (AFP) MACHINES

- Automated fiber placement technology has undergone significant improvements in the past decades. AFP process evolved to manufacture double-curved surfaces with the placement head automatically handling the start, stop, and cutting of individual tows. Additionally, automated fiber placement (AFP) machines have been improved by manufacturers by incorporating additional mechanisms such as gantry systems and robotic manipulators. The combination of accurate robot technologies and modular AFP heads provide a flexible platform that is capable of producing a wide variety of carbon fiber reinforced polymer (CFRP) parts with speed, quality, and reliability.
- For instance, Electroimpact Inc. has made several unique developments that improved the performance and reliability of automated fiber placement machines. Electroimpact Inc. offers accurate robot AFP system integrated with a large vacuum table, CNC controlled ultrasonic ply cutting gantry, and AFP head transfer stands. Some of the key characteristics of the AFP heads include simplified tension system, short, simple tow path, and high speed add on-the-fly and cut on-the-fly actuators. Electroimpact AFP machines have been used in commercial aircraft production and have demonstrated high rate, high quality production.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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PORTER'S FIVE FORCES ANALYSIS

Bargaining Power of Suppliers – LOW

- The AFP ATL machines market is characterized by presence of relatively high suppliers. Raw material suppliers tend to have long term contract with the companies.
- Therefore, suppliers have a low bargaining power in the AFP ATL machines market

Competitive Rivalry – HIGH

- The global automated fiber placements and automated tape laying (AFP- ATL) machines market shows high rivalry among players
- This can be attributed to several factors such as presence of big competitors that include Coriolis Composites, M.Torres Industrial Designs, SAU, Ingersoll Machine Tools Inc., and Electroimpact Inc. , among others and similar product offerings by these players.

Bargaining Power of Buyers – HIGH

- Buyers in the global AFP ATL machines market are limited and majorly aircraft manufacturers and research and development institutions
- Also, there is low switching cost. Therefore, the buyers' bargaining power in the AFP ATL machines market is high.

Threat of Substitutes – LOW

- There are relatively few substitutes for automated fiber placements and automated tape laying machines. Tape laying and fiber placement have become key technologies in the aircraft construction. Therefore, the threat of substitute in the AFP ATL machines market is low.

Threat of New Entrants – LOW

- Threat of new entrants in the global automated fiber placements and automated tape laying (AFP-ATL) machines market is low due to the high capital investment and R&D costs
- Additionally, presence of big players such as Coriolis Composites, Ingersoll Machine Tools Inc., and Electroimpact Inc. makes it difficult for new players to enter as they have established presence

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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TECHNOLOGY EVOLUTION IN AFP- ATL MARKET

- The first Automated Fiber Placement (AFP) machine was developed at the beginning of the 1980s and then commercialized in the 1990s.
- AFP was used as an R&D tool for many years before migrating to the production floor with the development of the Boeing 787 aircraft. AFP automatically places multiple individual prepreg tows onto a mandrel at high speed, using a numerically controlled placement head to dispense, clamp, cut and restart each tow during placement. Automated fiber placement (AFP) is emerging as one of the advanced methods of fabrication of polymer matrix based composite structures.
- Automated fiber placement technology has undergone significant technological developments in the past decades which include incorporation of additional mechanisms such as gantry systems and robotic manipulators. For instance, in 2016, M.Torres Industrial Designs, SAU delivered hybrid AFP/ATL gantry machines to Airbus for efficient A350XWB wings skins fabrication. This Gantry system is designed to deliver a high dynamic machine capable of extreme accelerations.
- For instance, Electroimpact Inc. has made several unique developments that improved the performance and reliability of automated fiber placement machines. Electroimpact Inc. offers the accurate robot AFP system integrated with a large vacuum table, CNC controlled ultrasonic ply cutting gantry, and AFP head transfer stands. Some of the key characteristics of the AFP heads include simplified tension system, short, simple tow path, and high speed add on-the-fly and cut on-the-fly actuators. Electroimpact AFP machines have been used in commercial aircraft production and have demonstrated high rate, high quality production.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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TECHNOLOGY EVOLUTION

- ATL machinery lays fiber down in the form of prepregged unidirectional tape or continuous strips of fabric, rather than single tows. Tape layup is versatile, allowing breaks in the process and easy changes in the direction of fiber orientation. ATL systems can also be adapted to process both thermoset and thermoplastic materials. For instance, Boikon B.V., a automation equipment providing company based in the Netherlands, offers Falko, an automated tape layer machine which can manufacture near-net-shape products and thermoplastic composite laminates. Furthermore, the equipment also can manufacture components up to 650 m2 per hour.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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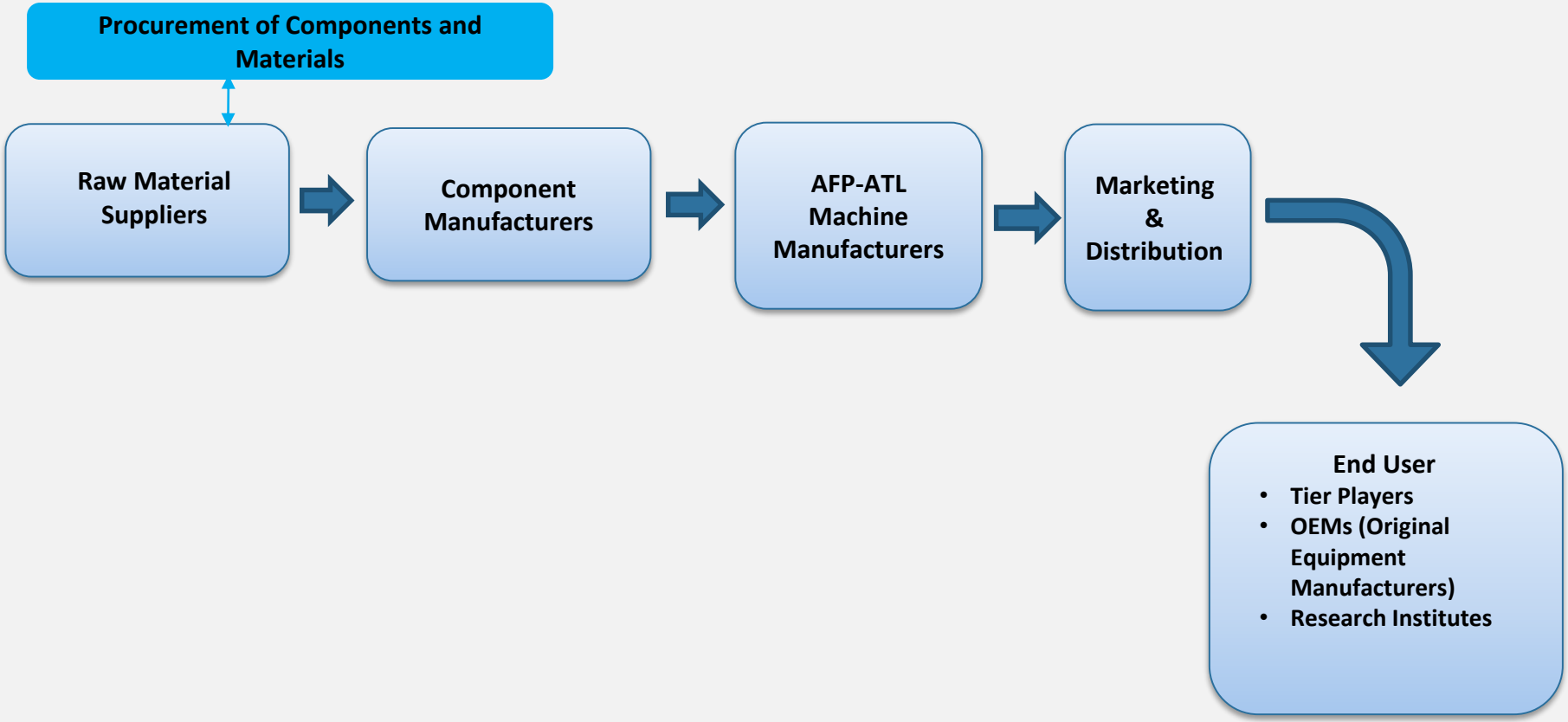
VALUE CHAIN ANALYSIS



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TRENDS IN WIND ENERGY INDUSTRY

- Automated Fiber Placements and Automated Tape Laying (AFP- ATL) machines offer completely automated production and industrial robustness due to the use of robotic components. Automated Tape Layup (ATL) or Automated Fiber Placement (AFP) machines are used in manufacturing wind turbine blades to reduce labor cost and improve quality.
- The enormous size of the wind turbine blades makes the manufacturing very challenging. Manufacturers in the wind energy industry are focusing on production of wind turbine blades by automating the layup of the composite fabrics used in the construction of wind turbine blades
- Coriolis Composites, a manufacturer of innovative equipment and software for the production of composite material structures, offers automated solutions for making wind turbine blades of large dimensions, optimizes the robot structure in relation to the components to be made, and enables the use of low-cost material, while ensuring high productivity.
- Moreover, Coriolis Composites offers C5 automated fiber placement (AFP) robotic gantry solution. Its multi-material capability enables AFP technology to be used for large structures such as wind turbines, marine applications, and civil engineering applications.

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TRENDS IN AEROSPACE INDUSTRY

- The aerospace industry is growing and manufacturers are searching for innovative ways to ramp up production. The airplanes are complex, highly engineered machines and their components must be manufactured with an extremely high degree of accuracy. Advancement in robotic integration solutions are helping the aerospace industry to overcome these challenges.
- The aircraft production process offers high potential for automation. Robotics can fill the needs of fabrication jobs including carbon fiber layup, trimming, routing, and drilling. For such fabrication process companies use Automated Fiber Placements and Automated Tape Laying (AFP- ATL) machines. Robots can also do assembly jobs such as fastening, sealing, and painting. Robots are also now used to inspect aircraft, a very time-intensive process due to strict standards required of the aircraft industry.
- AFP has become an important processing method and was identified as the most suitable to produce high performance aerospace components. MAGCincinnati, MTorres, Electroimpact, Coriolis and Broetje-Automation Composites are commercial manufacturers of AFP and ATL machines. AFP technology is currently being used in the aerospace industry to make parts up to 100 feet long.
- For instance, in November 2020, Firefly Aerospace announced to utilize Ingersoll Machine Tools' AFP systems for production of alpha rocket airframes. Firefly will install its first AFP system at its manufacturing and test facility in Texas, U.S. in May 2021, after which the Alpha rocket airframe will be requalified using AFP manufacturing processes. The second AFP system and automated assembly line is destined for Firefly's new Florida Space Coast factory and launch site at Cape Canaveral, set to be installed by the beginning of 2022.

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IMPACT OF COVID-19 PANDEMIC ON AFP- ATL MACHINES MARKET

- COVID-19 has significantly disrupted almost all the industries including, aerospace, automotive, manufacturing, wind energy, and others. To combat COVID-19 pandemic, most of the countries have implemented lockdown and are now easing it phase-wise and industry wise. This has significantly impacted manufacturing sector as the facilities across the globe were temporarily shut down.
- COVID-19 has turned into a global crisis and it has a significant impact on the supply-side as well as the demand side of the AFP and ATL machines market. Companies need to evaluate impacts on three fronts: supply chain, market demand, and workforce.
- The coronavirus pandemic has reduced commercial air travel significantly. The COVID-19 crisis has disrupted entire tier structure of the aerospace supply chain. The outbreak of COVID-19 has led to flight cancellations, travel bans and massive slow down in the supply chain activity.
- In commercial aviation, companies are experiencing disruption in production and slowing demand due to lack of availability of workers and travel bans. Demand for spare parts of aircraft is also down since less maintenance is required currently. Therefore, it is estimated to have negative impact on the growth of the global automated fiber placements and automated tape laying (AFP- ATL) machines market as the automated fiber placement (AFP) technology is mainly used in the aerospace industry. Aircraft manufacturers are capital intensive by nature, thus raising short-term concerns about cash flow and liquidity.
- Furthermore, aircraft manufacturers are cutting down the production capacities due to decreased demand for aircrafts and its components. This in turn is leading to decreasing demand for automated fiber placements and automated tape laying (AFP- ATL) machines in the aerospace industry.

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IMPACT OF COVID-19 PANDEMIC ON AFP- ATL MACHINES MARKET

- For instance, in August 2020, Boeing announced that it is reducing production of 787 aircraft from six aircraft per month to five aircraft per month by mid-2021. The 90% drop in international air travel in 2020, combined with the late-2020 outbreak in the COVID-19 pandemic, caused Boeing to scale down 787 aircraft production.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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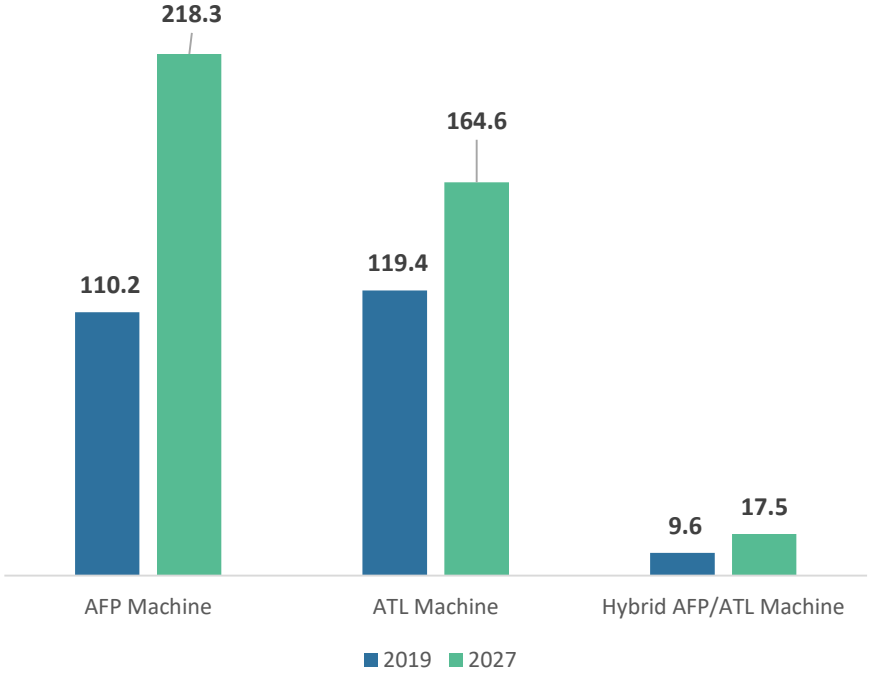
SECTION 4

GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINES MARKET, BY MACHINE TYPE

- AFP Machine
- ATL Machine
- Hybrid AFP/ATL Machine

AFP MACHINE SEGMENT IS EXPECTED TO EXHIBIT HIGH GROWTH OVER THE FORECAST PERIOD

FIGURE 4.1 Global Automated Fiber Placements and Automated Tape Laying (AFP- ATL) Machines Market Value (US$ Mn) Comparison, by Machine Type, 2019 & 2027



- On the basis of machine type, the global automated fiber placements and automated tape laying (AFP- ATL) machines market is segmented into AFP machine, ATL machine, and Hybrid AFP/ATL machine
- AFP machine segment is expected to account for largest share in 2027. This segment was valued at US$ 110.2 Mn in 2019 and is expected to generate US$ 218.3 Mn by 2027. This segment is expected to exhibit a CAGR of 9.6% over the forecast period from 2020 to 2027.
- Hybrid AFP/ATL machine segment was valued at US$ 9.6 Mn in 2019 and is expected to generate US$ 15.5 Mn by 2027. This segment is expected to exhibit CAGR of 7.7% over the forecast period from 2020 to 2027.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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MARKET VALUE, BY MACHINE TYPE

TABLE 4.1 Global Automated Fiber Placements and Automated Tape Laying (AFP- ATL) Machines Market Value (US$ Mn) Analysis and Forecast, By Machine Type, 2017 – 2027

Machine Type	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
AFP Machine	97.9	101.9	110.2	115.1	123.2	133.1	145.1	159.0	175.0	194.7	218.3	**9.6%**
ATL Machine	108.8	115.4	119.4	119.2	121.9	125.9	131.1	137.3	144.5	153.6	164.6	**4.7%**
Hybrid AFP/ATL Machine	7.3	8.5	9.6	10.4	11.3	12.3	13.3	14.3	15.4	16.4	17.5	**7.7%**
Total	**214.0**	**225.8**	**239.2**	**244.7**	**256.4**	**271.3**	**289.5**	**310.6**	**334.9**	**364.7**	**400.4**	**7.3%**

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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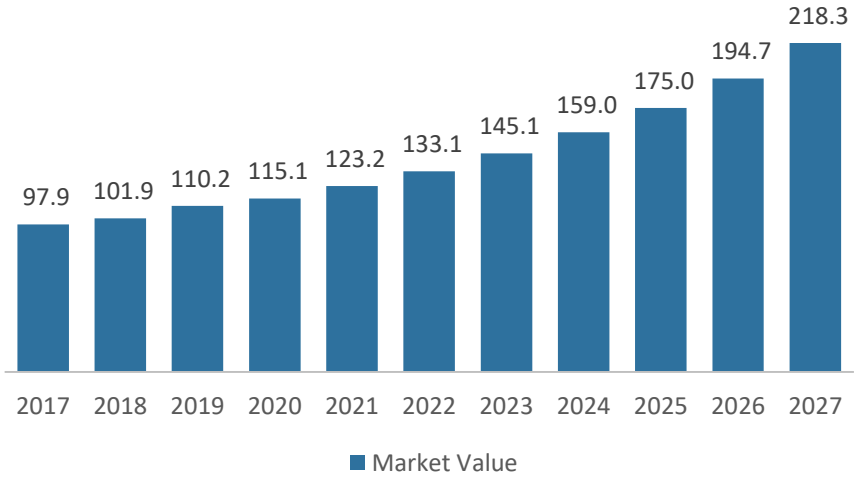
SECTION 4.1

GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINES MARKET, BY AFP MACHINE

- Segment Analysis

AFP MACHINE SEGMENT GENERATED REVENUE OF US$ 110.2 MN IN 2019

FIGURE 4.1.1 Global Automated Fiber Placements and Automated Tape Laying (AFP- ATL) Machines Market Value (US$ Mn) Analysis and Forecast, by AFP Machine



9.6% Projected growth rate of AFP machine segment between 2020 and 2027

- In 2019, the AFP machine segment generated revenue of US$ 110.2 Mn and is expected to reach US$ 218.3 Mn by 2027, exhibiting a CAGR of 9.6% over the forecast period 2020-2027

- AFP machine is defined as an automated fiber placement machine, which is an advanced method to fabricate composite structures for the aircraft's components such as airframes and other industrial products. AFP machine uses one ply (layer) at a time method, in order to build the complete structure. AFP provides several benefits such as fabrication of highly customized parts, as every single ply can be placed individually at different angles, making the component sustain more load compared to the conventional manufacturing method. Furthermore, other benefits offered by the AFP machine include shorter production lead time, faster lay-up rates, fewer stoppages and changeovers, increased material yield, and others. Due to these benefits AFP machines are gaining popularity in the aviation industry.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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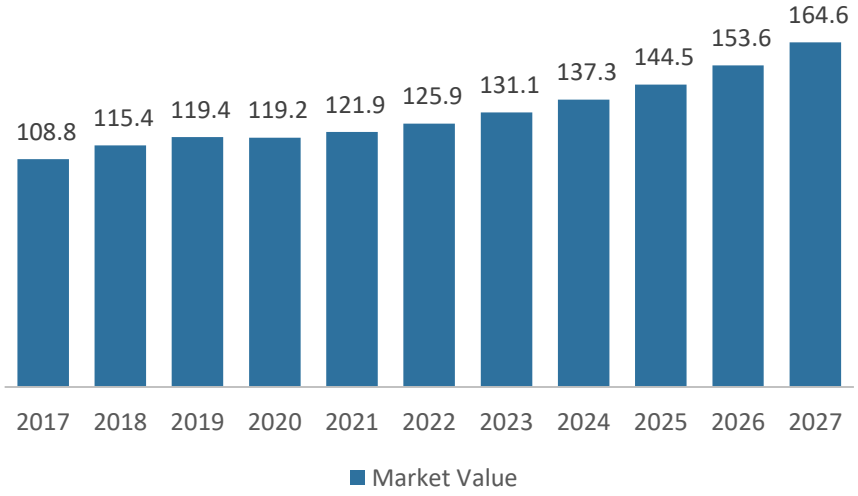
SECTION 4.2

GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINES MARKET, BY ATL MACHINE

- Segment Analysis

ATL MACHINE SEGMENT GENERATED REVENUE OF US$ 119.4 MN IN 2019

FIGURE 4.2.1 Global Automated Fiber Placements and Automated Tape Laying (AFP- ATL) Machines Market Value (US$ Mn) Analysis and Forecast, by ATL Machine



4.7% Projected growth rate of ATL machine segment between 2020 and 2027

- In 2019, the ATL machine segment generated revenue of US$ 119.4 Mn and is expected to reach US$ 164.6 Mn by 2027, exhibiting a CAGR of 4.7% over the forecast period 2020-2027

- ATL machines are also known as automated tape laying machine, which is used to manufacture thermoplastic composite laminates for the products of different shape and size using loaded roller system. The primary distinguishing factor between AFP and ATL machines is that AFP is more effective with material placing on contoured or curved surfaces, while ATL is more effective when a large amount of material is needed to be placed on minimally contoured surface or relatively large flat surfaces. ATL system has precise control of various tape operations including start, stop, cut, tape orientation, and others, allowing machines to fabricate complex reinforcement products.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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ATL MACHINE SEGMENT FORECAST

- Considering the increasing demand for the ATL machine, various key manufacturing companies are focusing on offering ATL machines for the manufacturing industry.
- For instance, Boikon B.V., a Netherland-based production automation equipment providing company, is offering Falko—an automated tape layer machine which can manufacture near-net-shape products and thermoplastic composite laminates. Furthermore, the equipment also can manufacture components up to 650 m2 per hour.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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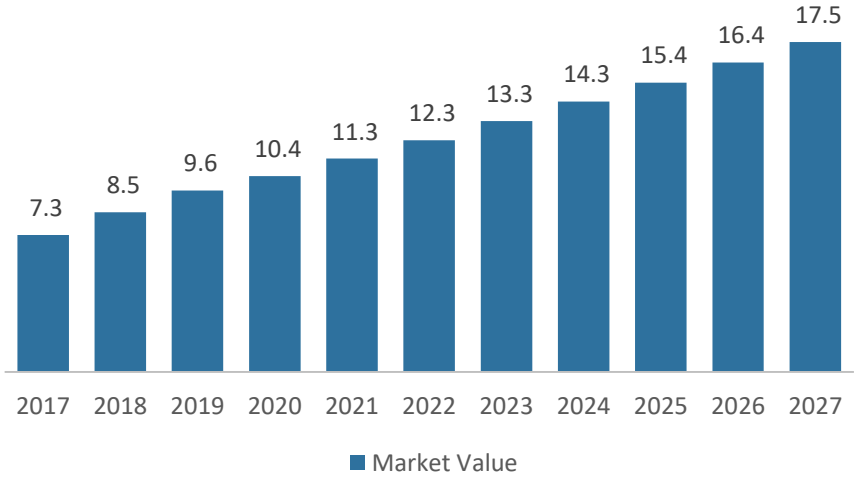
SECTION 4.3

GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINES MARKET, BY HYBRID AFP/ATL MACHINE

- Segment Analysis

HYBRID AFP/ATL MACHINE SEGMENT GENERATED REVENUE OF US$ 9.6 MN IN 2019

FIGURE 4.3.1 Global Automated Fiber Placements and Automated Tape Laying (AFP- ATL) Machines Market Value (US$ Mn) Analysis and Forecast, by Hybrid AFP/ATL Machine



7.7% Projected growth rate of Hybrid AFP/ATL machine segment between 2020 and 2027

- In 2019, the Hybrid AFP/ATL machine segment generated revenue of US$ 9.6 Mn and is expected to reach US$ 17.5 Mn by 2027, exhibiting a CAGR of 7.7% over the forecast period 2020-2027
- Hybrid AFP/ATL machine is a combination of both automated fiber placements and automated tape laying (AFP- ATL) machine. The primary aim of these machines is to increase the production efficiency of the component manufacturing cycle by reducing the production cycle time and reducing material waste.
- Due to increased production efficiency offered these machines, various key manufacturers are offering this equipment for the aviation industry. For instance, in October 2016, M.Torres Diseños Industriales SAU, a Spain-based equipment manufacturing company, offered hybrid AFP/ATL gantry machines to Airbus Group, a Netherland-based aircraft manufacturing company, in order to efficiently manufacture A350XWB wings skins.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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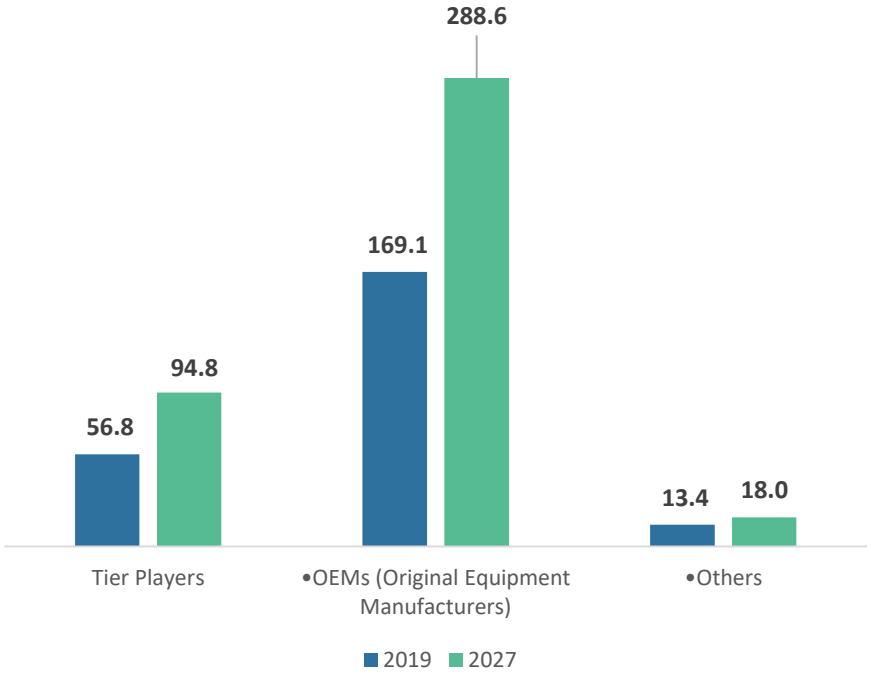
SECTION 5

GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP-ATL) MACHINES MARKET, BY END USER

- Tier Players
- OEMs (Original Equipment Manufacturers)
- Others

OEM SEGMENT IS EXPECTED TO EXHIBIT HIGH GROWTH DURING THE FORECAST PERIOD

FIGURE 5.1 Global Automated Fiber Placements and Automated Tape Laying (AFP-ATL) Machines Market Value (US$ Mn) Comparison, by End User, 2019 & 2027



- On the basis of end user, the global automated fiber placements and automated tape laying (AFP-ATL) machines market is segmented into Tier Players, OEMs (Original Equipment Manufacturers), and others
- OEMs (Original Equipment Manufacturers) segment accounted for the largest share in 2019. The segment was valued at US$ 169.1 Mn in 2019 and is expected to reach US$ 288.6 Mn by 2027. It is expected to exhibit a CAGR of 7.6% during the forecast period (2020 to 2027).
- Tier players segment was valued at US$ 56.8 Mn in 2019 and is expected to generate US$ 94.8 Mn by 2027.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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MARKET VALUE, BY END USER

TABLE 5.1 Global Automated Fiber Placements and Automated Tape Laying (AFP-ATL) Machines Market Value (US$ Mn) Analysis and Forecast, By End User, 2017 – 2027

End User	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
Tier Players	50.4	53.6	56.8	58.1	60.8	64.3	68.6	73.6	79.3	86.4	94.8	**7.3%**
OEMs (Original Equipment Manufacturers)	149.1	159.2	169.1	173.4	182.1	193.2	206.6	222.3	240.2	262.2	288.6	**7.6%**
Others	12.6	13.0	13.4	13.3	13.5	13.8	14.3	14.8	15.3	16.1	18.0	**4.4%**
Total	**212.0**	**225.8**	**239.2**	**244.7**	**256.4**	**271.3**	**289.5**	**310.6**	**334.9**	**364.7**	**401.4**	**7.3%**

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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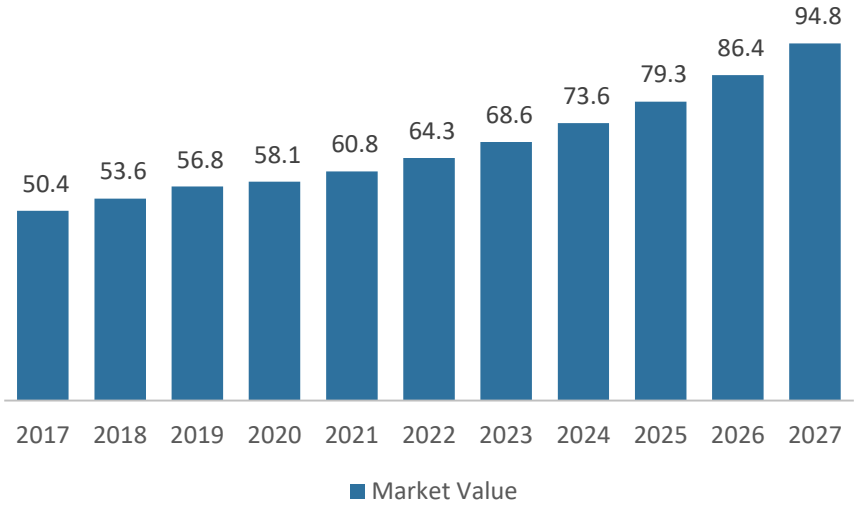
SECTION 5.1

GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP-ATL) MACHINES MARKET, BY TIER PLAYERS

- Segment Analysis

TIER PLAYERS SEGMENT GENERATED REVENUE OF US$ 56.8 MN IN 2019

FIGURE 5.1.1 **Global Automated Fiber Placements and Automated Tape Laying (AFP-ATL) Machines Market Value (US$ Mn) Analysis and Forecast, by Tier Players**



7.3% Projected growth rate of the tier players segment between 2020 and 2027

- In 2019, the tier players segment generated revenue of US$ 56.8 Mn and is expected to reach US$ 94.8 Mn by 2027, with a CAGR of 7.3% during the forecast period.
- Tier players companies in the aviation and automotive industry are focused on adoption of automated fiber placements and automated tape laying (AFP-ATL) machines. OEMs in the aviation industry are outsourcing various components manufacturing for aircraft and this is driving the market growth.
- For instance, in November 2020, Firefly Aerospace, a U.S. based private aerospace firm, announced that from 2021, the company will outsource all large part components from Ingersoll Machine Tools Inc., a U.S. based manufacturing company. All these components are made using AFP technology.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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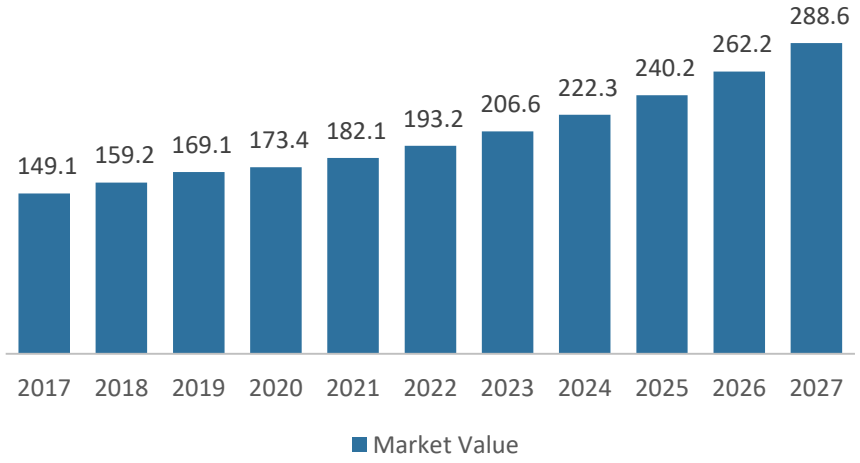
SECTION 5.2

GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP-ATL) MACHINES MARKET, BY OEMs (ORIGINAL EQUIPMENT MANUFACTURERS)

- Segment Analysis

OEMS (ORIGINAL EQUIPMENT MANUFACTURERS) SEGMENT GENERATED REVENUE OF US$ 169.1 MN IN 2019

FIGURE 5.2.1 **Global Automated Fiber Placements and Automated Tape Laying (AFP-ATL) Machines Market Value (US$ Mn) Analysis and Forecast, by OEMs (Original Equipment Manufacturers)**



7.6% Projected growth rate of the OEMs (original equipment manufacturers) segment between 2020 and 2027

- In 2019, the OEMs (original equipment manufacturers) segment generated revenue of US$ 169.1 Mn and is expected to reach US$ 288.6 Mn by 2027, with a CAGR of 7.6% during the forecast period

- The OEMs (original equipment manufacturers) segment has witnessed steady growth during the forecast period owing to the increasing demand for lightweight, sturdy components made using composite materials. For instance, according to Coherent Market Insights analysis, airplane manufacturing companies such as The Boeing Company and Airbus SE are planning to increase the use of AFP-ATL machines for Boeing 787 and Airbus A350 aircraft. By 2021, the companies are planning to increase their manufacturing capacity for these aircraft by around 80-100 per month; which is expected to facilitate the global automated fiber placements and automated tape laying (AFP-ATL) machines market growth.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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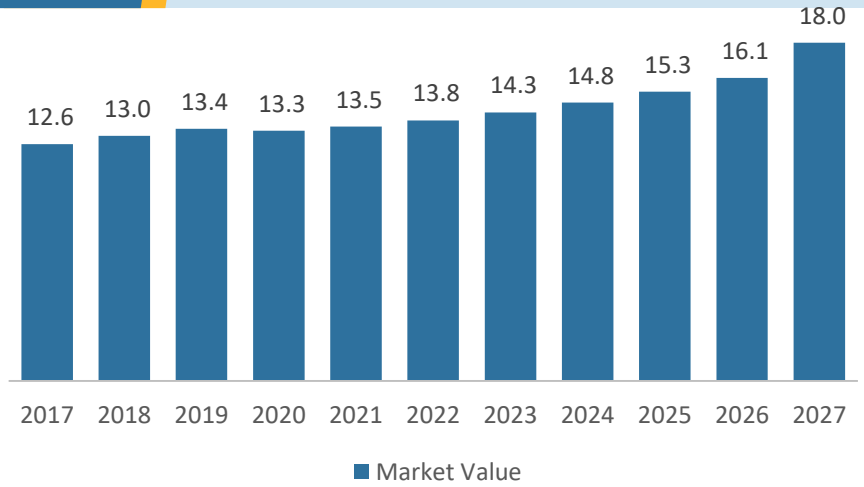
SECTION 5.3

GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP-ATL) MACHINES MARKET, BY OTHERS

- Segment Analysis

OTHERS SEGMENT GENERATED REVENUE OF US$ 13.4 MN IN 2019

FIGURE 5.3.1 Global Automated Fiber Placements and Automated Tape Laying (AFP-ATL) Machines Market Value (US$ Mn) Analysis and Forecast, by Others



4.4% Projected growth rate of the others segment between 2020 and 2027

- In 2019, the others segment generated revenue of US$ 13.4 Mn and is expected to reach US$ 18.0 Mn by 2027, with a CAGR of 4.4% during the forecast period
- The other segment for the AFP-ATL machines includes education and research institutes. Although AFP-ATL machines are rapidly being adopted in the aerospace industry, demand for AFP-ATL machines is still moderate in the automotive, wind energy, and other sectors due to high cost of AFP-ATL machines. Market key players are partnering with various educational institutes to reduce the overall cost and improve machine efficiency.
- For instance, in June 2020, Mikrosam DOO, A Europe-based AFP and ATL machine manufacturing company made a strategic partnership with the University of Delaware's Center for Composite Materials (UD-CCM) to develop an advanced ATP machines for automatic placement of dry, thermoplastic and thermoset tape.

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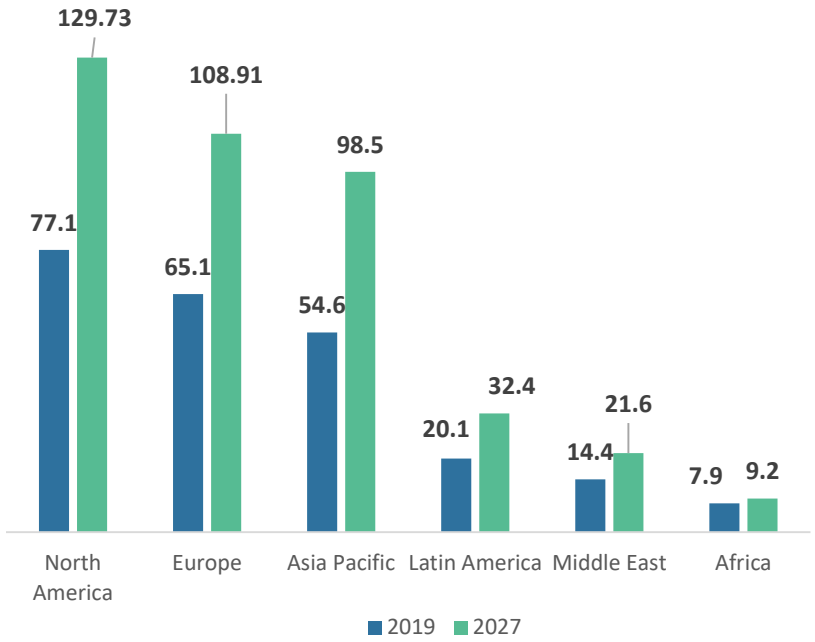
SECTION 6

GLOBAL AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP-ATL) MACHINES MARKET, BY REGION

- North America
- Europe
- Asia Pacific
- Latin America
- Middle East
- Africa

ASIA PACIFIC IS EXPECTED TO EXHIBIT HIGH GROWTH OVER THE FORECAST PERIOD

FIGURE 6.1 Global Automated Fiber Placements and Automated Tape Laying (AFP-ATL) Machines Market Value (US$ Mn) Comparison, By Region, 2019 & 2027



- On the basis of region, the global automated fiber placements and automated tape laying (AFP-ATL) machines market is segmented into North America, Europe, Asia Pacific, Latin America, Middle East, and Africa
- North America accounted for the largest share of 32.4% in 2019. The market was valued at US$ 77.1 Mn in 2019, and is expected to generate revenue of US$ 129.7 Mn by 2027. This market is expected to exhibit a CAGR of 7.4 % over the forecast period (2020 to 2027).
- Asia Pacific market is expected to exhibit the CAGR of 8.3% over the forecast period. This market was valued at US$ 54.6 Mn in 2019, and is expected to generate revenue of US$ 98.5 Mn by 2027

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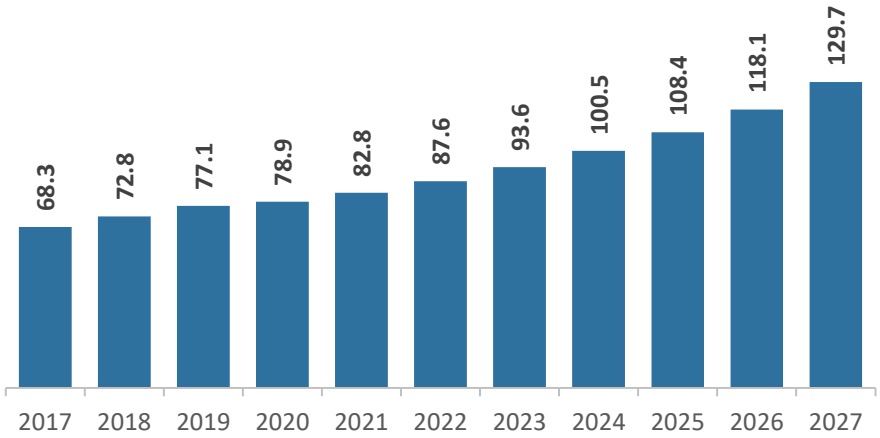
SECTION 6.1

NORTH AMERICA AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP-ATL) MACHINES MARKET

- Segmental Analysis

NORTH AMERICA GENERATED REVENUE OF US$ 77.1 MILLION IN 2019

Figure: 6.1.1 North America Automated Fiber Placements and Automated Tape Laying (AFP-ATL) Machines Market Value (US$ Mn) Analysis and Forecast



7.4% Projected growth rate of the North America market between 2020 and 2027

- North America market is expected to be valued at US$ 129.7 million by 2027 from US$ 77.1 million in 2019, witnessing a CAGR of 7.4% over the forecast period
- The demand for automated fiber placements and automated tape laying (AFP-ATL) machines is increasing in the North America region, and this is owing to rapidly growing aviation and space exploration sector in the U.S. Space exploration companies and government organizations such as The Boeing Company, The National Aeronautics and Space Administration (commonly known as NASA), and others are heavily investing in research and development for automated fiber placements and automated tape laying (AFP-ATL) machines, which will help them to reduce end-product cost drastically. Furthermore, according to a report published in August 2015, NASA is using composite material for its future space vehicles, which is developed in the composites Technology Center at NASA's Marshall Space Flight Center in Huntsville, Alabama. According to NASA, the newly developed composite

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- structure is made using fiber placements and automated tape laying (AFP-ATL) machines which offers higher strength and reduced overall production cost. Thus, the increasing demand by space exploration companies for composite materials will positively impact the North America Automated fiber placements and automated tape laying (AFP-ATL) machines market during the forecast period.

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NORTH AMERICA AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINES MARKET, BY COUNTRY

THE MARKET IN THE U.S. IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.1.1 North America Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machines Market Value (US$ MN) Analysis and Forecast, By Country, 2017 – 2027

COUNTRY	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
U.S.	54.2	57.9	61.5	63.0	66.2	70.2	75.1	80.8	87.3	95.3	105.0	**7.6%**
Canada	14.1	14.9	15.7	15.9	16.6	17.4	18.4	19.6	21.0	22.7	24.8	**6.5%**
Total	**68.3**	**72.8**	**77.1**	**78.9**	**82.8**	**87.6**	**93.6**	**100.5**	**108.4**	**118.1**	**129.7**	**7.4%**

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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AFP MACHINE SEGMENT IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.1.2 North America Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machines Market Value (US$ MN) Analysis and Forecast, By Machine Type, 2017 – 2027

MACHINE TYPE	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
AFP Machine	31.2	34.1	37.1	38.9	41.8	45.4	49.7	54.8	60.6	67.7	76.3	**10.1%**
ATL Machine	34.7	35.8	36.7	36.4	37.0	37.9	39.2	40.8	42.6	45.0	47.9	**4.0%**
Hybrid AFP/ATL Machine	2.4	2.9	3.3	3.6	3.9	4.3	4.6	4.9	5.2	5.4	5.6	**6.5%**
Total	**68.3**	**72.8**	**77.1**	**78.9**	**82.8**	**87.6**	**93.6**	**100.5**	**108.4**	**118.1**	**129.7**	**7.4%**

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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OEMs SEGMENT IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.1.3 North America Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machines Market Value (US$ MN) Analysis and Forecast, By End User, 2017 – 2027

End User	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
Tier Players	16.0	17.1	18.1	18.6	19.5	20.6	22.0	23.7	25.6	27.9	30.6	7.4%
OEMs (Original Equipment Manufacturers)	47.6	50.9	54.1	55.5	58.4	62.0	66.3	71.4	77.3	84.4	93.0	7.7%
Others	4.6	4.8	4.9	4.9	4.9	5.0	5.2	5.4	5.6	5.8	6.1	3.2%
Total	**68.3**	**72.8**	**77.1**	**78.9**	**82.8**	**87.6**	**93.6**	**100.5**	**108.4**	**118.1**	**129.7**	7.4%

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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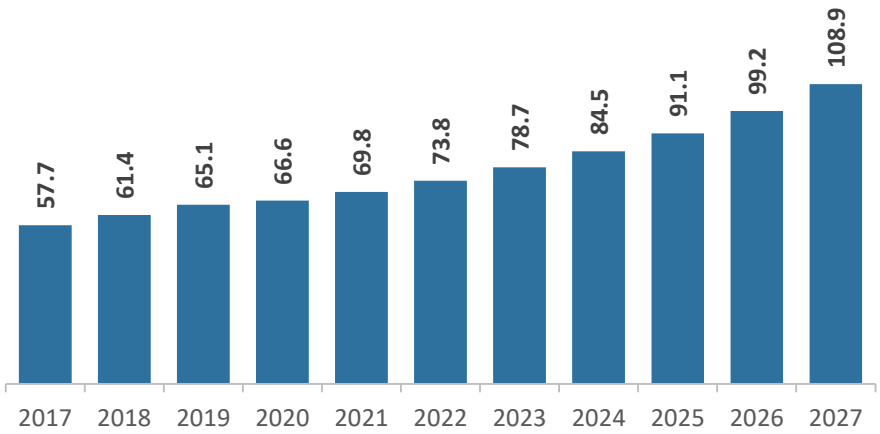
SECTION 6.2

EUROPE AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP-ATL) MACHINES MARKET

- Segmental Analysis

EUROPE GENERATED REVENUE OF US$ 65.1 MILLION IN 2019

Figure: 6.2.1 **Europe Automated Fiber Placements and Automated Tape Laying (AFP-ATL) Machines Market Value (US$ Mn) Analysis and Forecast**



7.3% Projected growth rate of the Europe market between 2020 and 2027

- Europe market is expected to be valued at US$ 108.9 million by 2027 from US$ 65.1 million in 2019, witnessing a CAGR of 7.3% over the forecast period
- The aviation industry in the Europe region exhibited high growth in the recent past. For instance, according to Coherent Market Insights' analysis, Europe commercial airlines witnessed rapid growth from 2013. In 2019, the number of passengers travelled through airlines in European Union reached almost 1.2 billion from just 0.75 billion in 2009. Furthermore, during the same time frame, the net profit for the commercial airlines also witnessed sharp growth. For instance, according to Eurostat, a government body for European statistics, in the span of 2013 to 2019, net profit for the airline companies reached almost US$ 6.5 billion in 2019 from almost 1 billion in 2013.
- Thus, considering the rapid growth in the aviation industry in Europe, various aircraft manufacturers are investing in advanced technologies such as automated fiber placements and automated tape laying (AFP-ATL) machines in order to increase their market share.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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EUROPE MARKET FORECAST

- For instance, in June 2015, M.Torres Diseños Industriales S.A.U, a Europe-based fiber placement machines is providing fiber placement machines to Airbus SE, a leading aircraft manufacturing company. Airbus is planning to use advanced technology from M.Torres Diseños Industriales S.A.U to manufacture Airbus’s 'Airbus 350 wings lamination' in the near future.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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EUROPE AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINES MARKET, BY COUNTRY

COHERENT MARKET INSIGHTS

GERMANY IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.2.1 Europe Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machines Market Value (US$ MN) Analysis and Forecast, By Country, 2017 – 2027

COUNTRY	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
UK	14.2	15.2	16.1	16.5	17.3	18.4	19.7	21.2	22.9	24.9	27.4	**7.5%**
France	11.4	12.2	12.9	13.2	13.9	14.7	15.7	16.8	18.2	19.8	21.8	**7.4%**
Germany	6.5	6.9	7.4	7.6	8.0	8.5	9.1	9.9	10.7	11.7	13.0	**7.9%**
Italy	4.6	4.8	5.0	5.1	5.3	5.5	5.8	6.2	6.6	7.1	7.7	**6.1%**
Russia	3.6	3.8	4.0	4.1	4.3	4.5	4.8	5.2	5.6	6.1	6.7	**7.1%**
Rest of Europe	17.4	18.5	19.6	20.0	20.9	22.1	23.6	25.2	27.1	29.5	32.3	**7.1%**
Total	**57.7**	**61.4**	**65.1**	**66.6**	**69.8**	**73.8**	**78.7**	**84.5**	**91.1**	**99.2**	**108.9**	**7.3%**

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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AFP MACHINE SEGMENT IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.2.2 Europe Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machines Market Value (US$ MN) Analysis and Forecast, By Machine Type, 2017 – 2027

MACHINE TYPE	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
AFP Machine	26.4	28.7	31.1	32.5	34.8	37.6	41.0	44.9	49.4	55.0	61.6	**9.6%**
ATL Machine	29.4	30.5	31.4	31.2	31.9	32.8	34.0	35.5	37.2	39.5	42.1	**4.4%**
Hybrid AFP/ATL Machine	1.8	2.2	2.6	2.8	3.1	3.4	3.8	4.1	4.4	4.8	5.1	**8.8%**
Total	**57.7**	**61.4**	**65.1**	**66.6**	**69.8**	**73.8**	**78.7**	**84.5**	**91.1**	**99.2**	**108.9**	**7.3%**

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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OEMs SEGMENT IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.2.3 Europe Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machines Market Value (US$ MN) Analysis and Forecast, By End User, 2017 – 2027

End User	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
Tier Players	13.8	14.7	15.5	15.8	16.6	17.5	18.7	20.0	21.5	23.4	25.7	**7.2%**
OEMs (Original Equipment Manufacturers)	41.1	43.8	46.5	47.7	50.1	53.2	56.9	61.2	66.1	72.1	79.4	**7.5%**
Others	2.8	2.9	3.0	3.0	3.0	3.1	3.2	3.3	3.4	3.6	3.8	**3.6%**
Total	**57.7**	**61.4**	**65.1**	**66.6**	**69.8**	**73.8**	**78.7**	**84.5**	**91.1**	**99.2**	**108.9**	**7.3%**

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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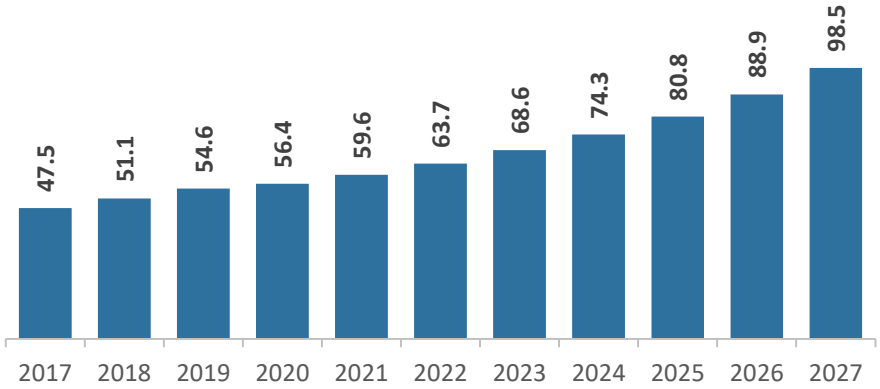
SECTION 6.3

ASIA PACIFIC AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP-ATL) MACHINES MARKET

- Segmental Analysis

ASIA PACIFIC GENERATED REVENUE OF US$ 54.6 MILLION IN 2019

Figure: 6.3.1 **Asia Pacific Automated Fiber Placements and Automated Tape Laying (AFP-ATL) Machines Market Value (US$ Mn) Analysis and Forecast**



8.3% Projected growth rate of the Asia Pacific market between 2020 and 2027

- Asia Pacific market is expected to be valued at US$ 98.5 million by 2027 from US$ 54.6 million in 2019, witnessing a CAGR of 8.3% over the forecast period
- Asia Pacific is considered one of the fastest growing regions in the global automated fiber placements and automated tape laying (AFP-ATL) machines market during the forecast period. This is owing to the increasing automation at industrial level coupled with various government policies in order to promote rapid industrialisation in the country.
- In emerging economies such as India and China, supply chain players plays a crucial role for the both automotive and aviation industries. For instance, according to Coherent Market Insights' analysis, China aircraft manufacturing industry is expected register CAGR of 5.6% during the period 2020-2027 which will aid the Asia Pacific automated fiber placements and automated tape laying (AFP-ATL) machines market growth.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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- Furthermore, government of India is also focusing on initiatives in order to promote domestic manufacturing and use of advanced technology such as automated fiber placements and automated tape laying (AFP-ATL) in the aviation industry. For instance, in May 2014, Government of India announced that it is planning to increase its domestic aircraft manufacturing, for which it allowed 100% foreign direct investment in the aviation sector.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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ASIA PACIFIC AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINES MARKET, BY COUNTRY

COHERENT MARKET INSIGHTS

INDIA IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.3.1 Asia Pacific Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machines Market Value (US$ MN) Analysis and Forecast, By Country, 2017 – 2027

COUNTRY	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
China	16.1	17.4	18.7	19.4	20.6	22.1	23.9	26.0	28.4	31.3	34.9	**8.7%**
India	9.0	9.8	10.6	11.1	11.8	12.8	13.9	15.2	16.7	18.6	20.8	**9.4%**
Australia	6.6	7.1	7.6	7.8	8.2	8.8	9.4	10.1	11.0	12.0	13.3	**7.9%**
Japan	4.7	5.1	5.5	5.7	6.0	6.5	7.0	7.6	8.3	9.1	10.1	**8.6%**
South Korea	4.7	5.0	5.3	5.5	5.8	6.1	6.6	7.1	7.7	8.4	9.3	**7.7%**
ASEAN	2.1	2.2	2.3	2.4	2.5	2.6	2.8	3.0	3.2	3.5	3.8	**7.0%**
Rest of APAC	4.1	4.3	4.5	4.5	4.7	4.8	5.1	5.3	5.6	5.9	6.3	**4.8%**
Total	**47.5**	**51.1**	**54.6**	**56.4**	**59.6**	**63.7**	**68.6**	**74.3**	**80.8**	**88.9**	**98.5**	**8.3%**

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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AFP MACHINE SEGMENT IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.3.2 Asia Pacific Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machines Market Value (US$ MN) Analysis and Forecast, By Machine Type, 2017 – 2027

MACHINE TYPE	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
AFP Machine	20.5	22.4	24.4	25.6	27.5	29.8	32.6	35.9	39.7	44.3	49.9	10.0%
ATL Machine	25.3	26.8	28.2	28.6	29.8	31.3	33.1	35.3	37.8	40.8	44.5	6.5%
Hybrid AFP/ATL Machine	1.7	1.9	2.1	2.2	2.4	2.6	2.9	3.1	3.4	3.7	4.0	8.9%
Total	**47.5**	**51.1**	**54.6**	**56.4**	**59.6**	**63.7**	**68.6**	**74.3**	**80.8**	**88.9**	**98.5**	8.3%

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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OEMs SEGMENT IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.3.3 Asia Pacific Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machines Market Value (US$ MN) Analysis and Forecast, By End User, 2017 – 2027

End User	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
Tier Players	11.2	12.1	12.9	13.3	14.1	15.1	16.3	17.6	19.2	21.1	23.4	**8.4%**
OEMs (Original Equipment Manufacturers)	33.2	35.8	38.4	39.8	42.2	45.2	48.8	53.0	57.8	63.7	70.8	**8.6%**
Others	3.0	3.2	3.3	3.3	3.3	3.4	3.5	3.7	3.8	4.0	5.2	**7.0%**
Total	**47.5**	**51.1**	**54.6**	**56.4**	**59.6**	**63.7**	**68.6**	**74.3**	**80.8**	**88.9**	**99.5**	**8.5%**

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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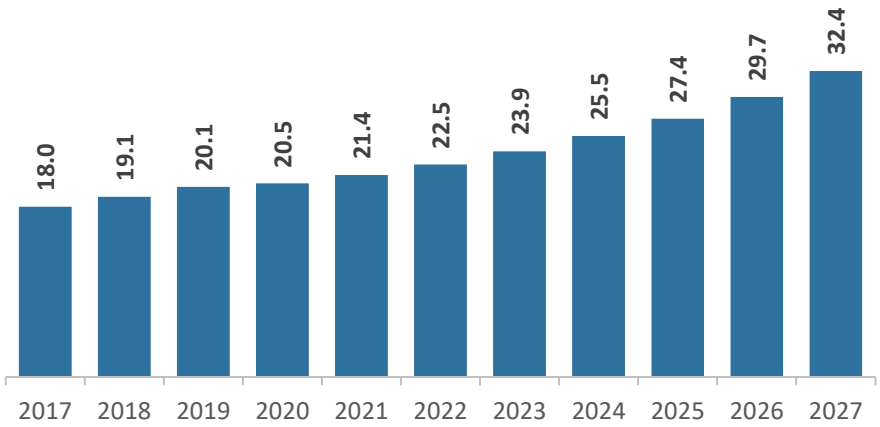
SECTION 6.4

LATIN AMERICA AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP-ATL) MACHINES MARKET

- Segmental Analysis

LATIN AMERICA SEGMENT GENERATED REVENUE OF US$ 20.1 MILLION IN 2019

Figure: 6.4.1 **Latin America Automated Fiber Placements and Automated Tape Laying (AFP-ATL) Machines Market Value (US$ Mn) Analysis and Forecast**



6.8% Projected growth rate of the Latin America market between 2020 and 2027

- Latin America market is expected to be valued at US$ 32.4 million by 2027 from US$ 20.1 million in 2019, witnessing a CAGR of 6.8% over the forecast period
- Latin America region is expected to witness steady growth over the forecast period. This is due to the rapidly growing automotive sector in the region. For instance, according to Coherent Market Insights' analysis, Latin America automotive sales reached 2.88 million units in 2018 from 2.70 million units in 2017. The growing automotive sector in the Latin America region will positively aid in the Latin America automated fiber placements and automated tape laying (AFP-ATL) machines market growth during the forecast period.
- Composite materials play a crucial role in the vehicle part manufacturing as it provides high strength-to-weight ratio for automotive vehicles and enables manufacturers to provide better fuel efficiency, lower overall weight, and cost. Furthermore, two-step AFP processing enables automotive manufacturers to increase the overall production rate of composite.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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BRAZIL IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.4.1 Latin America Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machine Market Value (US$ MN) Analysis and Forecast, By Country, 2017 – 2027

COUNTRY	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
Mexico	4.7	5.0	5.2	5.3	5.6	5.9	6.2	6.7	7.2	7.8	8.5	**6.9%**
Brazil	7.5	8.0	8.4	8.6	9.0	9.4	10.0	10.7	11.5	12.5	13.7	**7.0%**
Argentina	1.9	2.0	2.1	2.1	2.2	2.3	2.5	2.6	2.8	3.1	3.3	**6.6%**
Rest of Latin America	3.9	4.2	4.4	4.4	4.6	4.9	5.1	5.5	5.8	6.3	6.9	**6.4%**
Total	**18.0**	**19.1**	**20.1**	**20.5**	**21.4**	**22.5**	**23.9**	**25.5**	**27.4**	**29.7**	**32.4**	**6.8%**

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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AFP MACHINES SEGMENT IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.4.2 Latin America Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machines Market Value (US$ MN) Analysis and Forecast, By Machine Type, 2017 – 2027

MACHINE TYPE	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
AFP Machine	7.7	8.3	8.9	9.3	9.8	10.5	11.4	12.4	13.5	14.9	16.5	**8.6%**
ATL Machine	9.7	10.1	10.4	10.4	10.7	11.1	11.5	12.1	12.7	13.5	14.5	**4.8%**
Hybrid AFP/ATL Machine	0.6	0.7	0.7	0.8	0.9	0.9	1.0	1.1	1.2	1.3	1.4	**7.9%**
Total	**18.0**	**19.1**	**20.1**	**20.5**	**21.4**	**22.5**	**23.9**	**25.5**	**27.4**	**29.7**	**32.4**	**6.8%**

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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OEMs SEGMENT IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.4.3 Latin America Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machines Market Value (US$ MN) Analysis and Forecast, By End User, 2017 – 2027

End User	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
Tier Players	4.3	4.6	4.8	4.9	5.1	5.3	5.6	6.0	6.4	6.9	7.6	**6.5%**
OEMs (Original Equipment Manufacturers)	12.8	13.5	14.3	14.6	15.2	16.1	17.1	18.3	19.6	21.3	23.3	**6.9%**
Others	0.9	1.0	1.0	1.0	1.1	1.1	1.2	1.3	1.3	1.4	1.6	**5.9%**
Total	**18.0**	**19.1**	**20.1**	**20.5**	**21.4**	**22.5**	**23.9**	**25.5**	**27.4**	**29.7**	**32.4**	**6.8%**

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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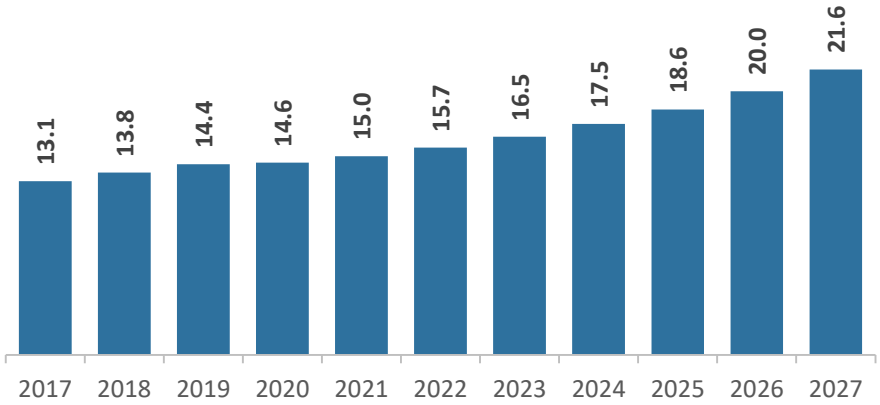
SECTION 6.5

MIDDLE EAST AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP-ATL) MACHINES MARKET

- Segmental Analysis

MIDDLE EAST SEGMENT GENERATED REVENUE OF US$ 14.4 MILLION IN 2019

Figure: 6.5.1 **Middle East Automated Fiber Placements and Automated Tape Laying (AFP-ATL) Machines Market Value (US$ Mn) Analysis and Forecast**



5.8% Projected growth rate of the Middle East market between 2020 and 2027

- Middle East market is expected to be valued at US$ 21.6 million by 2027 from US$ 14.4 million in 2019, witnessing a CAGR of 5.8% over the forecast period
- Middle East automated fiber placements and automated tape laying (AFP-ATL) machines market is expected to witness steady growth over the forecast period. This growth is owing to the increasing demand for composite material in the oil and gas industry.
- Carbon fiber composite components made using AFP-ATL machines can sustain in ultra-deepwater environments where low temperatures, extreme water pressure, and saltwater corrosion can cause serious damage to the equipment made using non-composite materials and metals
- Moreover, the oil and gas industry in the Middle East region is expected to grow at a rapid pace with the increasing government and foreign investments in the region.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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MIDDLE EAST MARKET FORECAST

- For instance, according to CMI's analysis in 2018, China (including the Chinese government and private companies) invested around US$ 80 billion in the Middle East oil and gas projects to support its The Belt and Road Initiative.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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MIDDLE EAST AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINE MARKET, BY COUNTRY

COHERENT MARKET INSIGHTS

GCC COUNTRIES IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.5.1 Middle East Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machine Market Value (US$ MN) Analysis and Forecast, By Country, 2017 – 2027

COUNTRY	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
GCC Countries	8.3	8.8	9.2	9.3	9.6	10.1	10.7	11.3	12.1	13.0	14.1	**6.1%**
Israel	0.7	0.8	0.8	0.8	0.8	0.8	0.8	0.9	0.9	1.0	1.0	**3.8%**
Rest of Middle East	4.1	4.3	4.4	4.5	4.6	4.8	5.0	5.3	5.6	6.0	6.5	**5.5%**
Total	**13.1**	**13.8**	**14.4**	**14.6**	**15.0**	**15.7**	**16.5**	**17.5**	**18.6**	**20.0**	**21.6**	**5.8%**

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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AFP MACHINES SEGMENT IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.5.2 Middle East Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machines Market Value (US$ MN) Analysis and Forecast, By Machine Type, 2017 – 2027

MACHINE TYPE	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
AFP Machine	7.4	5.5	5.9	6.0	6.3	6.7	7.2	7.7	8.3	9.1	10.0	**7.6%**
ATL Machine	5.2	7.7	7.9	7.9	8.0	8.2	8.5	8.9	9.3	9.9	10.5	**4.3%**
Hybrid AFP/ATL Machine	0.6	0.6	0.7	0.7	0.7	0.8	0.8	0.9	0.9	1.0	1.1	**6.8%**
Total	**13.1**	**13.8**	**14.4**	**14.6**	**15.0**	**15.7**	**16.5**	**17.5**	**18.6**	**20.0**	**21.6**	**5.8%**

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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OEMs SEGMENT IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.5.3 Middle East Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machines Market Value (US$ MN) Analysis and Forecast, By End User, 2017 – 2027

End User	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
Tier Players	3.2	3.3	3.5	3.5	3.7	3.8	4.0	4.3	4.5	4.9	5.3	**5.9%**
OEMs (Original Equipment Manufacturers)	9.3	9.7	10.2	10.3	10.7	11.2	11.8	12.5	13.3	14.3	15.5	**6.0%**
Others	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.8	0.8	0.8	**2.0%**
Total	**13.1**	**13.8**	**14.4**	**14.6**	**15.0**	**15.7**	**16.5**	**17.5**	**18.6**	**20.0**	**21.6**	**5.8%**

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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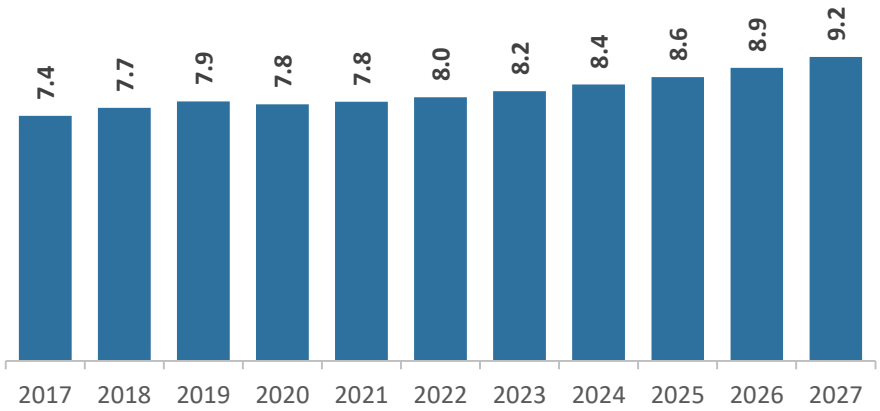
SECTION 6.6

AFRICA AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP-ATL) MACHINES MARKET

- Segmental Analysis

AFRICA SEGMENT GENERATED REVENUE OF US$ 7.9 MILLION IN 2019

Figure: 6.6.1 **Africa Automated Fiber Placements and Automated Tape Laying (AFP-ATL) Machines Market Value (US$ Mn) Analysis and Forecast**



2.5% Projected growth rate of the Africa market between 2020 and 2027

- Africa market is expected to be valued at US$ 9.2 million by 2027 from US$ 7.9 million in 2019, witnessing a CAGR of 2.5% over the forecast period
- Africa's automated fiber placements and automated tape laying (AFP-ATL) machines market is expected to grow due to the rapidly increasing wind energy market. Adoption of AFP-ATL machines is increasing in the wind blade manufacturing as it enables manufacturers to improve quality and reduce labor costs.
- However, the wind energy market in the Africa region is small and concentrated in a few parts of the region. Factors such as large global investment and government initiatives to provide clean energy to people is driving the market growth. For instance, according to Global Wind Energy Council (GWEC), a Belgium-based non-profit organization, in Africa, in August 2017, the total power generation capacity using wind energy in the Africa region was recorded 4.1GW.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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AFRICA MARKET FORECAST

- According to the International Energy Agency, a Paris-based autonomous intergovernmental organization, the total power generation capacity using wind energy in Africa region is expected to reach 385GW by 2040. Thus, the rapidly growing wind energy market in the Africa region will positively aid in growth of the Africa's automated fiber placements and automated tape laying (AFP-ATL) machines market over the forecast period.

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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AFRICA AUTOMATED FIBER PLACEMENTS AND AUTOMATED TAPE LAYING (AFP- ATL) MACHINE MARKET, BY COUNTRY

COHERENT MARKET INSIGHTS

SOUTH AFRICA IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.6.1 Africa Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machine Market Value (US$ MN) Analysis and Forecast, By Country, 2017 – 2027

COUNTRY	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
South Africa	2.46	2.56	2.64	2.62	2.66	2.72	2.79	2.88	2.97	3.08	3.21	**3.0%**
Rest of Africa	4.96	5.11	5.23	5.15	5.19	5.27	5.37	5.49	5.63	5.80	6.00	**2.2%**
Total	**7.4**	**7.7**	**7.9**	**7.8**	**7.8**	**8.0**	**8.2**	**8.4**	**8.6**	**8.9**	**9.2**	**2.5%**

Source: Secondary and Primary Research, Coherent Market Insights, 2019-20

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AFP MACHINES SEGMENT IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.6.2 Africa Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machines Market Value (US$ MN) Analysis and Forecast, By Machine Type, 2017 – 2027

MACHINE TYPE	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
AFP Machine	2.7	2.8	2.9	2.9	3.0	3.1	3.2	3.4	3.5	3.7	3.9	**4.1%**
ATL Machine	4.5	4.6	4.7	4.6	4.6	4.6	4.7	4.7	4.8	4.9	5.0	**1.3%**
Hybrid AFP/ATL Machine	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	**3.2%**
Total	**7.4**	**7.7**	**7.9**	**7.8**	**7.8**	**8.0**	**8.2**	**8.4**	**8.6**	**8.9**	**9.2**	**2.5%**

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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OEMs SEGMENT IS EXPECTED TO WITNESS HIGHEST GROWTH OVER THE FORECAST PERIOD

TABLE 6.6.3 Africa Automated Fiber Placements And Automated Tape Laying (AFP- ATL) Machines Market Value (US$ MN) Analysis and Forecast, By End User, 2017 – 2027

End User	2017 H	2018 H	2019 B	2020 E	2021 F	2022 F	2023 F	2024 F	2025 F	2026 F	2027 F	CAGR (2020–2027)
Tier Players	1.8	1.9	1.9	1.9	1.9	1.9	2.0	2.0	2.0	2.1	2.2	**2.2%**
OEMs (Original Equipment Manufacturers)	5.2	5.4	5.5	5.5	5.5	5.7	5.8	6.0	6.1	6.3	6.6	**2.7%**
Others	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	**1.1%**
Total	**7.4**	**7.7**	**7.9**	**7.8**	**7.8**	**8.0**	**8.2**	**8.4**	**8.6**	**8.9**	**9.2**	**2.5%**

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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SECTION 7

COMPETITIVE LANDSCAPE

- Company Profiles

Ingersoll Machine Tools, Inc.

Company Name

Ingersoll Machine Tools, Inc.

Establishment

1891

Headquartered

Illinois, U.S.

Key Management

Mr. Chip Storie (CEO)

Employees (2019)

~180

Website

www.ingersoll.com

Revenue (2019)

~ 51.9

Company Overview

- Ingersoll Machine Tools, Inc. develops, manufactures, and supplies machine tools and equipment
- The company offers rotor slots, scalpers, locomotives, wind blades, horizontal milling machines, and engine blocks
- The offerings of the company are segmented into various key areas such as additive manufacturing, fiber, composite manufacturing, vertical machines, horizontal machines, and special machines.
- The company conducts its business worldwide

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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PRODUCT PORTFOLIO

PRODUCT PORTFOLIO

- **Composite Manufacturing**
  - Mongoose
  - Mongoose Hybrid
  - Robotic FP
  - Hawk

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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KEY DEVELOPMENTS

2020

On December 11, Ingersoll Machine Tools Inc., was awarded as the 'Manufacturer of the Year' by the Rockford Chamber of Commerce in a virtual event and ceremony presented by Illinois Bank & Trust.

2020

In October, Firefly Aerospace, Inc., one of the leading provider of economical launch vehicles, spacecraft, and in-space services, announced a substantial commitment to increase its manufacturing capacity by transitioning large parts manufacturing to automated fiber placement mongoose hybrid systems from Ingersoll machine tools in the beginning of 2021.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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KEY BUSINESS STRATEGIES

PARTNERSHIP

- **The company is focused on strategies such as partnerships, in order to expand its market share**

- In May 2012, Ingersoll Machine Tools, Inc. entered into partnership with Lockheed Martin to develop a new and visibly improved software package for the programming of automated fiber placement (AFP) equipment. Lockheed Martin is a U.S.-based aerospace, defense, arms, security, and advanced technologies company.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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Automated Dynamics

Company Name

Automated Dynamics

Establishment

1984

Headquartered

Niskayuna, U.S.

Key Management

Mr. Robert Langone (General Manager)

Employees (2019)

~20-50

Website

www.automateddynamics.com

Revenue (2019)

~7.1

Company Overview

- Automated Dynamics designs automation equipment for composite part production
- The company offers right-sized, automated fiber placement (AFP) and automated tape laying (ATL) equipment for thermoset and thermoplastic applications
- The company serves various sectors such as aerospace, automotive, construction, defense, industrial, life sciences, oil & gas, and research
- Trelleborg Sealing Solutions Germany GmbH is a parent company of Automated Dynamics. In 2017, Trelleborg Sealing Solutions fully acquired U.S.-based privately owned Automated Dynamics.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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PRODUCT PORTFOLIO

PRODUCT PORTFOLIO

- Fiber Placement
- Tape Laying
- Research Workcell
- Software

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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Automated Dynamics

KEY DEVELOPMENTS

2013

In June, Automated Dynamics announced the sale of its state-of-the-art Automated Fiber Placement (AFP) system to the Long Island Forum for Technology (LIFT), Bethpage, New York.

2011

In November, Automated Dynamics launched AFP machine for R&D work.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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KEY BUSINESS STRATEGIES

INNOVATION

- **The company is focused on strategies such as innovation in fiber placement and tape laying, in order to expand its market share**
  - In April 2015, Automated Dynamics issued patent, which covers methods to automate incorporation of lighting strike protection layers into composite structures. This development helps to lower production costs throughout several industries, where such protection is necessary, including but not limited to aerospace, oil and gas, wind energy, and others.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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M.Torres Industrial Designs, SAU

Company Name
M.Torres Industrial Designs, SAU

Establishment
1975

Headquartered
Navarra, Spain

Key Management
Mr. Manuel Torres (Founder)

Employees (2019)
501-1,000

Website
www.mtorres.com

Revenue (2019)
~113.43

Company Overview

- M.Torres Industrial Designs, SAU offers systems for industrial processes automation and solutions for the energy and environmental sector
- The company offers aerospace industry machinery, machine tools for manufacturing aerospace components, tooling systems, and jigs and assembly lines for machines
- Currently the company has provided its products in over 70 countries with over 620 customers

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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PRODUCT PORTFOLIO

PRODUCT PORTFOLIO

- ATL Machine
- AFP machine
- Hybrid AFP/ATL Gantry Machines

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KEY DEVELOPMENTS

2019

In March, the company provided one Robotic AFP to each aerospace companies such as Leonardo in Italy and COMAC in China, for their own R&D activities related to lamination of carbon fiber material, both with thermoset as well as thermoplastic capabilities.

2018

In March, the company received the JEC Innovation Award for the new automated manufacturing process without mold for monocoque fuselages in composite material.

2016

In March, the company delivered hybrid AFP/ATL gantry machines to Airbus for efficient A350XWB wings skins fabrication

2015

In September, the company delivered automatic tape layer (ATL) machine to Boeing aerospace company. This was 6th Automatic Tape Layer machine of Boeing.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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KEY BUSINESS STRATEGIES

PRODUCT DEVELOPMENT

- **The company is focused on strategies such as product development, in order to expand its market share**
  - In February 2017, the company launched new 2D Integrated Torreslayup System, which combines highly reliable and most productive ATL head, and other beneficial optional components, functions and elements
  - In May 2015, the company started manufacturing of the fifth AFP MTorres machine for A350 XWB Wing Spars in GKN Western Approach plant, Bristol (U.K.)
  - In June 2015, the company provided fiber placement machines to Airbus, which are used at Stade (Germany) and Illescas (Spain) plants to manufacture its latest A350 model upper and lower wing skins

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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Electroimpact Inc.

Company Name

Electroimpact Inc.

Establishment

1986

Headquartered

Washington, U.S.

Key Management

Mr. Peter Zieve (CEO)

Employees (2019)

201-500

Website

www.electroimpact.com

Company Overview

- Electroimpact Inc. designs and manufactures aerospace tooling and automation solutions
- The company offers automated fiber placement technology, automated fastening systems, drilling machines, hand held electromagnetic rivet guns, aerospace ground support equipment, and robotic systems.
- The company's wide range of projects include complete automation assembly systems for commercial aircraft wings, riveting machines and tools for wing panel and fuselage assembly, advanced fiber placement machines, robotic assembly systems, and spacecraft handling equipment.
- The company serves the aircraft industry worldwide

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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PRODUCT PORTFOLIO

PRODUCT PORTFOLIO

- **Composites Manufacturing**
  - AFP 4.0

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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KEY DEVELOPMENTS

2019

In April, Electroimpact Inc. provided a new robotic automated fiber placement (AFP) system to Spirit AeroSystems, featuring an upgraded AFP head said to place composite material more than twice as fast and nearly three times more reliably than traditional AFP heads

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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KEY BUSINESS STRATEGIES

PRODUCT DEVELOPMENT

- **The company is focused on strategies such as product development, in order to expand its market share**

- In October 2020, Electroimpact Inc. launched 6-axis printer that combines AFP and fused filament fabrication (FFF) technologies. Electroimpact Inc. combined two processes such as automated fiber placement (AFP) and fused filament fabrication (FFF) 3D printing into a unified system called Scalable Composite Robotic Additive Manufacturing. The 6-axis SCRAM allows tool-free fabrication of aerospace-grade structures from continuous-fiber-reinforced (CFR) composites.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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Mikrosam

Company Name

Mikrosam

Establishment

1990

Headquartered

Prilep, North Macedonia

Key Management

Ms. Liljana Samak (Co-Founder)

Employees (2019)

~80

Website

mikrosam.com

Company Overview

- Mikrosam offers automated fiber and tape placement (AFP/ATL), prepreg making machines, prepreg slitting/rewinding equipment, filament winding (FW) machines, and composite machining
- The company serves various industries such as aerospace, automotive & road transportation, electrical, electronics, telecom & appliances, equipment and machinery, marine transportation & shipbuilding, and renewable energy

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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PRODUCT PORTFOLIO

PRODUCT PORTFOLIO

- **Automated Fiber and Tape Placement (AFP/ATL)**
  - Automated Fiber Placement (AFP)
  - Automated Tape Laying (ATL)
  - AFP/ATL Robotic Cell
  - Offline Programing Software
  - Online Motion and Process Control

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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KEY DEVELOPMENTS

2019

In December, Mikrosam was selected by GKN Fokker to deliver advanced equipment for development of aircraft parts with thermoplastic composites (TPC). This include Mikrosam's latest automated fiber placement (AFP) machine, slitting equipment for thermoplastic prepregs, and a custom-engineered thermoplastic consolidating machine.

2018

In December, Mikrosam delivered multi-material capable automated fiber placement (AFP) system to the Department of Lightweight Design and Structural Materials (LsW) at Brandenburg University of Technology Cottbus. The 8-axis robotic AFP machine is capable of automated and precise placement of thermoplastic fibres, thermoset prepregs, and dry-fibre material.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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KEY BUSINESS STRATEGIES

PARTNERSHIP

- **The company is focused on strategies such as partnerships, in order to expand its market share**
  - In November 2017, Mikrosam entered into partnership with Composite Automation LLC to bring its cutting-edge solutions to the North American Market. With the growth in advanced composite parts production, the need for fiber placement and tape laying (AFP/ATL) systems in the North American market has never increased.

PRODUCT LAUNCH

- **The company is focused on strategies such as product launches, in order to expand its market share**
  - In February 2017, Mikrosam launched new AFP/ATL robot solution machine that integrates AFP head for thermoset and ATL head for thermoplastic prepreg laying into a single robotic cell for custom development of structural composites

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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Accudyne Systems, Inc

Company Name

Accudyne Systems, Inc

Establishment

1996

Headquartered

Newark, U.S.

Key Management

Mr. Ralph Cope (President/Owner)

Employees (2019)

~11-50

Website

www.accudyne.com

Company Overview

- Accudyne Systems, Inc. offers custom automation equipment
- The offerings of company are segmented into three different areas such as composite automation, supercritical fluid processing, and industrial automation
- The company offers project concept through design, fabrication, assembly, testing, installation, and support services
- Composite automation segment include automation assisted work cells, automated fiber placement (AFP), automated tape laying (ATL), forming/compaction/debulking, composite pick and place, cross ply and fabric lamination, and specialty machines

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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PRODUCT PORTFOLIO

PRODUCT PORTFOLIO

- **Composites Automation**
  - **Automation Fiber Placement (AFP)**
    - AFP I
    - AFP II
    - AFPIII
  - **Automated Tape Laying (ATL)**
    - Automated Tape Laminator
    - Ply-by-Ply Tape Laminator
    - Thermoplastic Staker / Laminator
    - Electron Beam Tape Laminator
    - Insitu Thermoplastic Laminator I
    - Insitu Thermoplastic Laminator II
    - Lab-Scale ATL Machine

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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Fives

Company Name

Fives

Establishment

1812

Headquartered

Paris, France

Key Management

Mr. Frederic Sanchez (CEO)

Employees (2019)

~ 8,400

Website

www.fivesgroup.com

Revenue (2019)

US$ 2,441.48 Mn

Company Overview

- Fives is a company, which designs and supplies machine tool and complete manufacturing solutions in a broad range of industrial sectors
- Products of the company are segmented into broaching & sawing, composites, drilling, laser applications, milling, and turning
- In composites segment, the company provides, fiber placement systems, tape layer systems, fishing/trimming, and controls & software
- The company serves various industries such as, aerospace, agriculture, automotive / truck, construction, job shop / general machining, mining, marine, oil/gas, power generation, rail and wind

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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PRODUCT PORTFOLIO

PRODUCT PORTFOLIO

- **Composites**
  - **Fiber Placement Systems**
    - Cincinnati VIPER FPS
    - Cincinnati GEMINI Dockable Gantry System
  - **Tape Layer Systems**
    - Cincinnati CHARGER Tape Layers
    - Cincinnati GEMINI Dockable Gantry System
    - Forest-Line ATLAS One
    - Forest-Line ATLAS
    - Forest-Line ACCESS
  - **Controls & Software**
    - Controls & Software
    - Forest-Line TapeLay

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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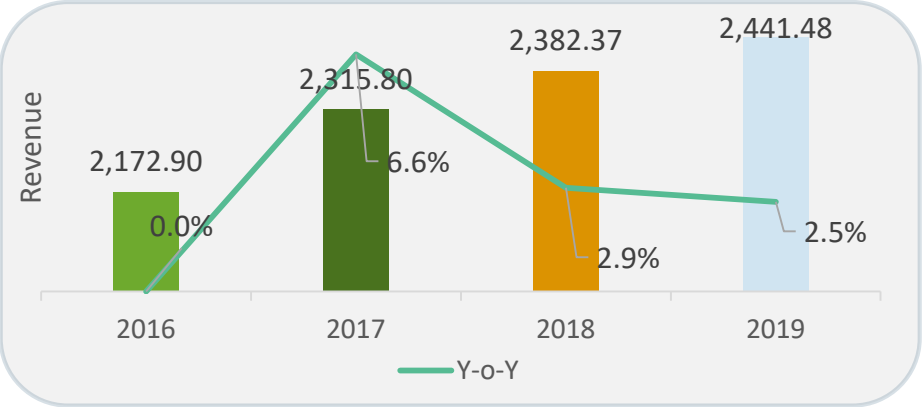
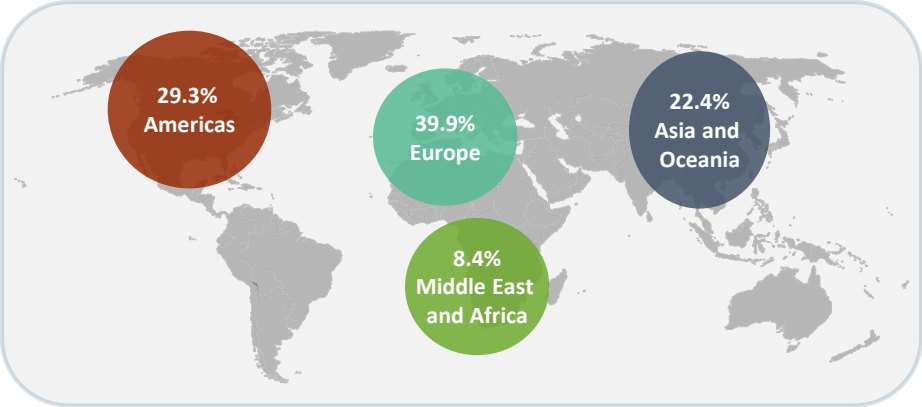
REVENUE (US$ Mn) & Y-o-Y ANALYSIS (2016-2019)



REVENUE SPLIT (%), BY REGION/COUNTRY, 2019



Year	2017	2018	2019
Revenue (US$ Mn)	2,315.80	2,382.37	2,441.48

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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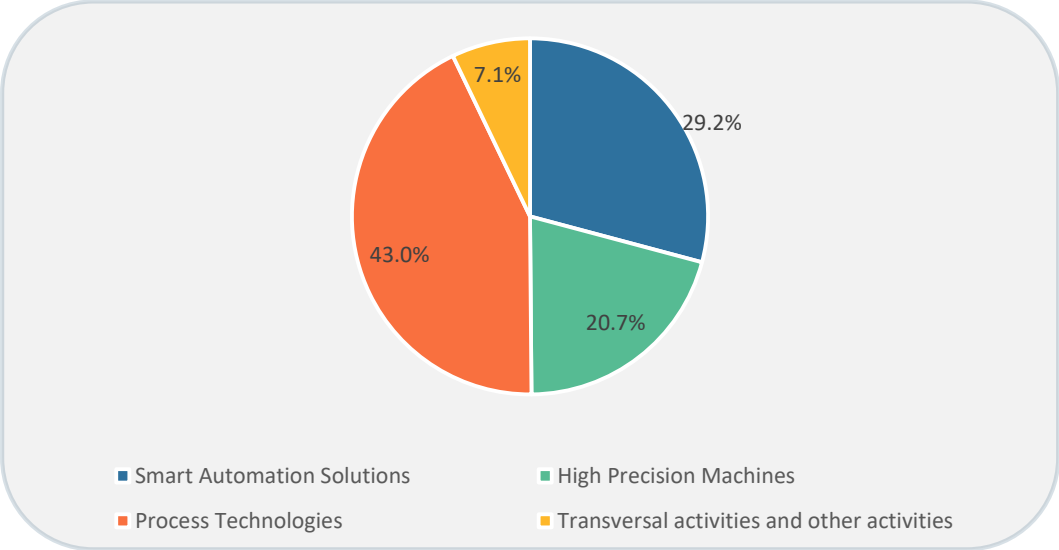
REVENUE SPLIT (%), BY SEGMENTS, 2019



Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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KEY BUSINESS STRATEGIES

ACQUISITION

- **The company is focused on strategies such as acquisition in order to expand its market share**
- In October 2020, Fives acquired assets and activities from Dufieux, a high-tech machine-tool designer and manufacturer

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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Coriolis Composites

Company Name

Coriolis Composites

Establishment

2000

Headquartered

Queven, France

Key Management

Mr. Clementine Gallet (CEO)

Employees (2019)

~170

Website

www.coriolis-composites.com

Company Overview

- Coriolis Composites is a company which develops, produces and commercializes robotic cells and value-added software for automated composite additive manufacturing.
- Coriolis Composites is one of the leading suppliers of automated fiber placement machines. The company offers its solutions in a number of industrial sectors such as aerospace, automotive and renewable energies.
- The company has offices in Augsburg (Germany), Bristol (U.K), Montreal (Canada), Seattle (U.S.A.) and Shanghai (China).
- Coriolis Group is a parent company of Coriolis Composites.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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PRODUCT PORTFOLIO

PRODUCT PORTFOLIO

- Fiber Placement Machines
- Composite Engineering
- Software Solutions

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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KEY DEVELOPMENTS

2018

In October, Coriolis Composites was selected by the National Institute for Aviation Research (NIAR) at Wichita State University to provide a thermoplastics capable Automated Fiber Placement (AFP) system.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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KEY BUSINESS STRATEGIES

PRODUCT DEVELOPMENT

- **The company is focused on strategies such as product development in order to expand its market share**
  - In March 2018, Coriolis Composites developed robotic systems for automated fiber placement (AFP) using low-cost, high-accuracy six-axis robots for high throughput.

PARTNERSHIP

- **The company is focused on strategies such as partnerships to strengthen its market position**
  - In May 2020, Coriolis Composites demonstrated inline inspection in partnership with Edixia Automation (Vern-sur-Seiche, France), which supplies automated inspection and industrial vision technologies. The sensor built by Edixia Automation is integrated into the placement head of Coriolis Composite's AFP machines.

Source: Primary and Secondary Research, Coherent Market Insights, 2019-20

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RESEARCH METHODOLOGY

RESEARCH PROCESS

Our research is an optimum culmination of secondary and primary research, with the latter playing a major role. We also have an in-house repository and access to a number of external regional and global paid databases to help address specific requirements of our clients. The research process begins with a holistic secondary research, followed by expert interviews. The information gathered is then comprehensively analyzed by our highly experienced in-house research panel.

PROPOSAL & MARKET TAXONOMY	MARKET ANALYSIS	MARKET EVALUATION & FORECASTING	DATA VERIFICATION AND VALIDATION	SERVICE/REPORT LAUNCH
▪ Market Scope ▪ Technical Analysis ▪ Market Taxonomy ▪ Prospect Database	▪ Future Growth Prospects ▪ Market Trend Analysis ▪ Competitive Landscape ▪ Absolute dollar opportunity ▪ Ecosystem Analysis ▪ Regional Trends and Analysis	▪ Historic Data ▪ Current Market Size ▪ Future Market Potential	▪ Primary Research ▪ Data Standardization ▪ Expert Panel Interviews ▪ Response Analysis ▪ Holistic analysis using Data Triangulation Methodology	▪ Press Release ▪ Whitepaper ▪ Social Media ▪ Campaigns Consultation
• Pilot Prospect		• Reconsideration • Value Addition		

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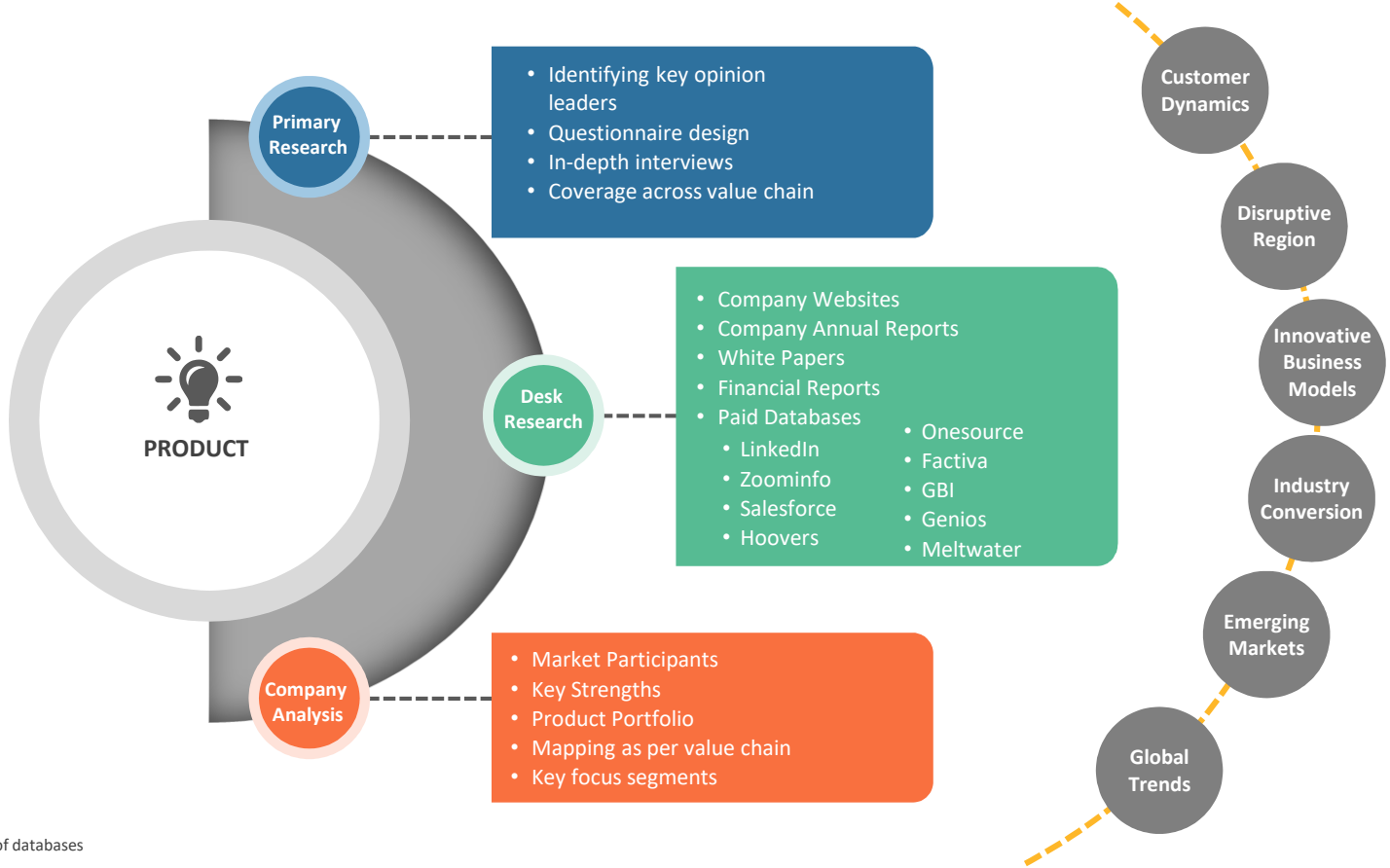
RESEARCH APPROACH



includes sources of databases

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RESEARCH APPROACH (DATA ANALYSIS AND ESTIMATION)

GLOBAL AFP- ATL MACHINES MARKET

OPTIMISTIC

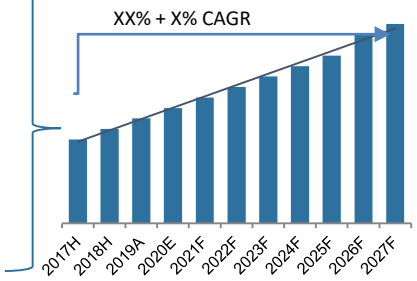
- Market indicators expected to outperform during the forecast period
- Market expected to gain traction with increasing impact of drivers

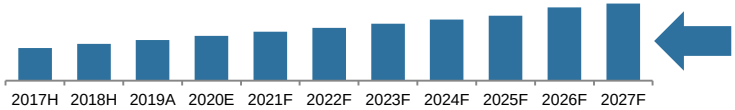
LIKELY

- Market indicators expected to outperform during the forecast period
- Market expected to gain traction with increasing impact of drivers

	SHORT TERM	MEDIUM TERM	LONG TERM
D1			
D2			
D3			

Degree of impact High Medium Low

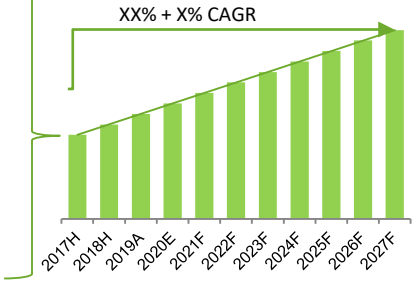




- Qualitative to quantitative conversion
- Developing the actual forecast based on both the cases

	SHORT TERM	MEDIUM TERM	LONG TERM
D1			
D2			
D3			

Degree of impact High Medium Low



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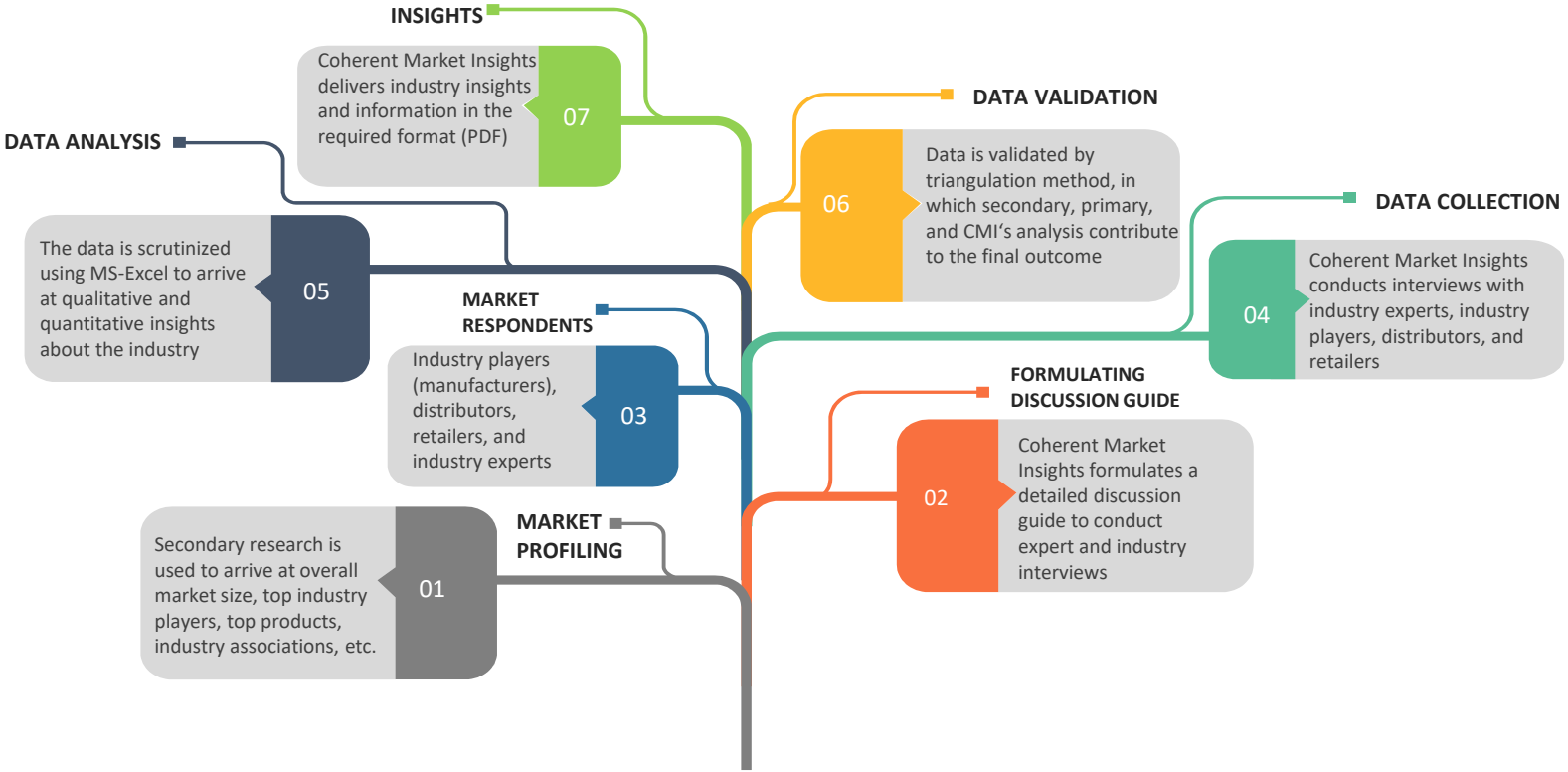

SYSTEMATIC RESEARCH APPROACH

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ABOUT US AND SERVICES

ABOUT Coherent Market Insights

Coherent Market Insights is a global market intelligence and consulting organization focused on assisting its plethora of clients achieve transformational growth by helping them make critical business decisions. We are headquartered in India, with an office at the global financial capital in the U.S. Our client base includes players from across all business verticals in over 150 countries worldwide. We are uniquely positioned to help businesses around the globe deliver practical and lasting results through various recommendations about operational improvements, technologies, emerging market trends and new working methods. We offer both customized and syndicated market research reports that help our clients create visionary growth plans to provide traction to their business. We meticulously study emerging trends across various industries at both the global and regional levels to identify new opportunities for our clientele.

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Analyzing multiple parameters for market research ensures a high level of conformance with expert opinions of the industry

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- Pricing Analysis
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- Impact of IoT Assessment
- Industry Assessment
- Regulations
- Service and Product Manufacturing Assessment
- Manufacturing Plants Assessment
- Energy Efficient Systems Assessment

- SWOT Analysis
- Ecosystem Analysis
- Supply Chain Analysis
- Porter's Analysis
- PESTLE Analysis
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- Patents Analysis
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